AF-400 INVERTER DRIVES

PARALLELING DRIVER - 331X441AAG01

OPERATION – TROUBLE SHOOTING – REPAIR

GEK-62516

DRIVER ASSEMBLY	ELEM	36D870002AA
	CONN	36B590247AA

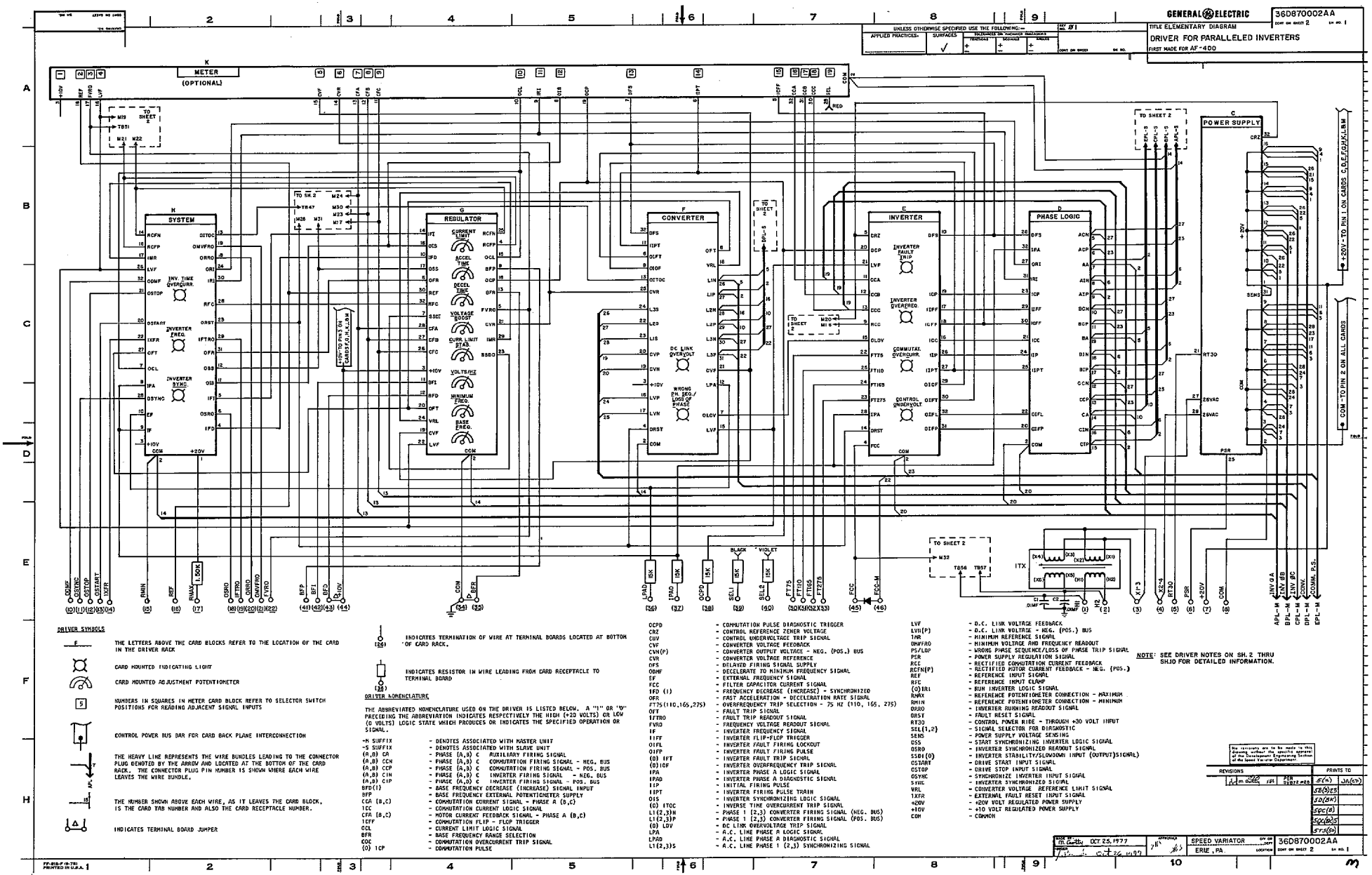
PRINTED CIRCUIT CARD DIAGRAMS

PHASE LOGIC	36D868807AA
INVERTER	36D868810AA
CONVERTER	36D868812AA
REGULATOR	36D868816AA
SYSTEM	36D868818AA
POWER SUPPLY	36C764164AA
	36D868820AA
METER	36D868822AA
CURRENT FEEDBACK	36C764166AA
PULSE TRANSFORMER	36C764172AA
PARALLELING	36D868831AA

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

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DRIVER SYMBOLS

F — THE LETTERS ABOVE THE CARD BLOCKS REFER TO THE LOCATION OF THE CARD IN THE DRIVER RACK

CARD MOUNTED INDICATING LIGHT

CARD MOUNTED ADJUSTMENT POTENTIOMETER

5 — NUMBERS IN SQUARES IN METER CARD BLOCK REFER TO SELECTOR SWITCH POSITIONS FOR READING ADJACENT SIGNAL INPUTS

CONTROL POWER BUS BAR FOR CARD BACK PLANE INTERCONNECTION

THE HEAVY LINE REPRESENTS THE WIRE BUNDLES LEADING TO THE CONNECTOR PLUG DENOTED BY THE ARROW AND LOCATED AT THE BOTTOM OF THE CARD RACK. THE CONNECTOR PLUG PIN NUMBER IS SHOWN WHERE EACH WIRE LEAVES THE WIRE BUNDLE.

15 — THE NUMBER SHOWN ABOVE EACH WIRE, AS IT LEAVES THE CARD BLOCK, IS THE CARD TAB NUMBER AND ALSO THE CARD RECEPTACLE NUMBER.

INDICATES TERMINAL BOARD JUMPER

(26) — INDICATES TERMINATION OF WIRE AT TERMINAL BOARDS LOCATED AT BOTTOM OF CARD RACK.

(38) — INDICATES RESISTOR IN WIRE LEADING FROM CARD RECEPTACLE TO TERMINAL BOARD

DRIVER NOMENCLATURE

THE ABBREVIATED NOMENCLATURE USED ON THE DRIVER IS LISTED BELOW. A "1" OR "0" PRECEDING THE ABBREVIATION INDICATES RESPECTIVELY THE HIGH (+20 VOLTS) OR LOW (0 VOLTS) LOGIC STATE WHICH PRODUCES OR INDICATES THE SPECIFIED OPERATION OR SIGNAL.

-H SUFFIX	- DENOTES ASSOCIATED WITH MASTER UNIT
-S SUFFIX	- DENOTES ASSOCIATED WITH SLAVE UNIT
(A,B) CA	- PHASE (A,B) C AUXILIARY FIRING SIGNAL
(A,B) CCN	- PHASE (A,B) C COMMUTATION FIRING SIGNAL - NEG. BUS
(A,B) CCP	- PHASE (A,B) C COMMUTATION FIRING SIGNAL - POS. BUS
(A,B) CIN	- PHASE (A,B) C INVERTER FIRING SIGNAL - NEG. BUS
(A,B) CIP	- PHASE (A,B) C INVERTER FIRING SIGNAL - POS. BUS
BFD(I)	- BASE FREQUENCY DECREASE (INCREASE) SIGNAL INPUT
BFP	- BASE FREQUENCY EXTERNAL POTENTIOMETER SUPPLY
CCA (B,C)	- COMMUTATION CURRENT SIGNAL - PHASE A (B,C)
CCL	- COMMUTATION CURRENT LOGIC SIGNAL
CFA (B,C)	- MOTOR CURRENT FEEDBACK SIGNAL - PHASE A (B,C)
CFF	- COMMUTATION FLIP - FLOP TRIGGER
CGL	- CURRENT LIMIT LOGIC SIGNAL
CFR	- BASE FREQUENCY RANGE SELECTION
COC	- COMMUTATION OVERCURRENT TRIP SIGNAL
(0) ICP	- COMMUTATION PULSE
OCPD	- COMMUTATION PULSE DIAGNOSTIC TRIGGER
CRZ	- CONTROL REFERENCE ZENER VOLTAGE
CUV	- CONTROL UNDERVOLTAGE TRIP SIGNAL
CVF	- CONVERTER VOLTAGE FEEDBACK
CVN(P)	- CONVERTER OUTPUT VOLTAGE - NEG. (POS.) BUS
CVR	- CONVERTER VOLTAGE REFERENCE
DFS	- DELAYED FIRING SIGNAL SUPPLY
DDMF	- DECELERATE TO MINIMUM FREQUENCY SIGNAL
EF	- EXTERNAL FREQUENCY SIGNAL
FCC	- FILTER CAPACITOR CURRENT SIGNAL
IFD (I)	- FREQUENCY DECREASE (INCREASE) - SYNCHRONIZED
OFR	- FAST ACCELERATION - DECELERATION RATE SIGNAL
FT75(110,165,275)	- OVERFREQUENCY TRIP SELECTION - 75 HZ (110, 165, 275)
OFT	- FAULT TRIP SIGNAL
IFTRO	- FAULT TRIP READOUT SIGNAL
FVRO	- FREQUENCY VOLTAGE READOUT SIGNAL
IF	- INVERTER FREQUENCY SIGNAL
IIFF	- INVERTER FLIP-FLOP TRIGGER
OIFL	- INVERTER FAULT FIRING LOCKOUT
OIFP	- INVERTER FAULT FIRING PULSE
(0) IFT	- INVERTER FAULT TRIP SIGNAL
(0)IOF	- INVERTER OVERFREQUENCY TRIP SIGNAL
IPA	- INVERTER PHASE A LOGIC SIGNAL
IPAD	- INVERTER PHASE A DIAGNOSTIC SIGNAL
IIP	- INITIAL FIRING PULSE
IIPT	- INVERTER FIRING PULSE TRAIN
OIS	- INVERTER SYNCHRONIZING LOGIC SIGNAL
(0) ITOC	- INVERSE TIME OVERCURRENT TRIP SIGNAL
L1(2,3)N	- PHASE 1 (2,3) CONVERTER FIRING SIGNAL (NEG. BUS)
L1(2,3)P	- PHASE 1 (2,3) CONVERTER FIRING SIGNAL (POS. BUS)
(0) LOV	- DC LINK OVERVOLTAGE TRIP SIGNAL
LPA	- A.C. LINE PHASE A LOGIC SIGNAL
LPAD	- A.C. LINE PHASE A DIAGNOSTIC SIGNAL
L1(2,3)S	- A.C. LINE PHASE 1 (2,3) SYNCHRONIZING SIGNAL
LVF	- D.C. LINK VOLTAGE FEEDBACK
LVR(P)	- D.C. LINK VOLTAGE - NEG. (POS.) BUS
MR	- MINIMUM REFERENCE SIGNAL
OMVFRO	- MINIMUM VOLTAGE AND FREQUENCY READOUT
PS/LOP	- WRONG PHASE SEQUENCE/LOSS OF PHASE TRIP SIGNAL
PSR	- POWER SUPPLY REGULATION SIGNAL
RCC	- RECTIFIED COMMUTATION CURRENT FEEDBACK
RCFM(P)	- RECTIFIED MOTOR CURRENT FEEDBACK - NEG. (POS.)
REF	- REFERENCE INPUT SIGNAL
RFC	- REFERENCE INPUT CLAMP
(0)IRI	- RUN INVERTER LOGIC SIGNAL
RMAX	- REFERENCE POTENTIOMETER CONNECTION - MAXIMUM
RMIN	- REFERENCE POTENTIOMETER CONNECTION - MINIMUM
ORRO	- INVERTER RUNNING READOUT SIGNAL
ORST	- FAULT RESET SIGNAL
RT30	- CONTROL POWER RIDE - THROUGH +30 VOLT INPUT
SEL(1,2)	- SIGNAL SELECTOR FOR DIAGNOSTIC
SENS	- POWER SUPPLY VOLTAGE SENSING
OSS	- START SYNCHRONIZING INVERTER LOGIC SIGNAL
OSRO	- INVERTER SYNCHRONIZED READOUT SIGNAL
SSDI(O)	- INVERTER STABILITY/SLOWDOWN INPUT (OUTPUT)SIGNAL)
OSTART	- DRIVE START INPUT SIGNAL
OSTOP	- DRIVE STOP INPUT SIGNAL
OSYNC	- SYNCHRONIZE INVERTER INPUT SIGNAL
SYNC	- INVERTER SYNCHRONIZED SIGNAL
VRL	- CONVERTER VOLTAGE REFERENCE LIMIT SIGNAL
IXFR	- EXTERNAL FAULT RESET INPUT SIGNAL
+20V	- +20V VOLT REGULATED POWER SUPPLY
+10V	- +10 VOLT REGULATED POWER SUPPLY
COM	- COMMON

NOTE: SEE DRIVER NOTES ON SH. 2 THRU SH.10 FOR DETAILED INFORMATION.

GENERAL ELECTRIC — 36D870002AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES: — SURFACES: ✓ — TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± — DECIMALS ± — ANGLES ±

TITLE ELEMENTARY DIAGRAM

DRIVER FOR PARALLELED INVERTERS

FIRST MADE FOR AF-400

REVISIONS		PRINTS TO
Add 0220	181 PER SU072-028	5(A) JA(C)
		5B(B)23
		5D(D)K
		5PC(B)
		5QC(B)5
		5TS(B)

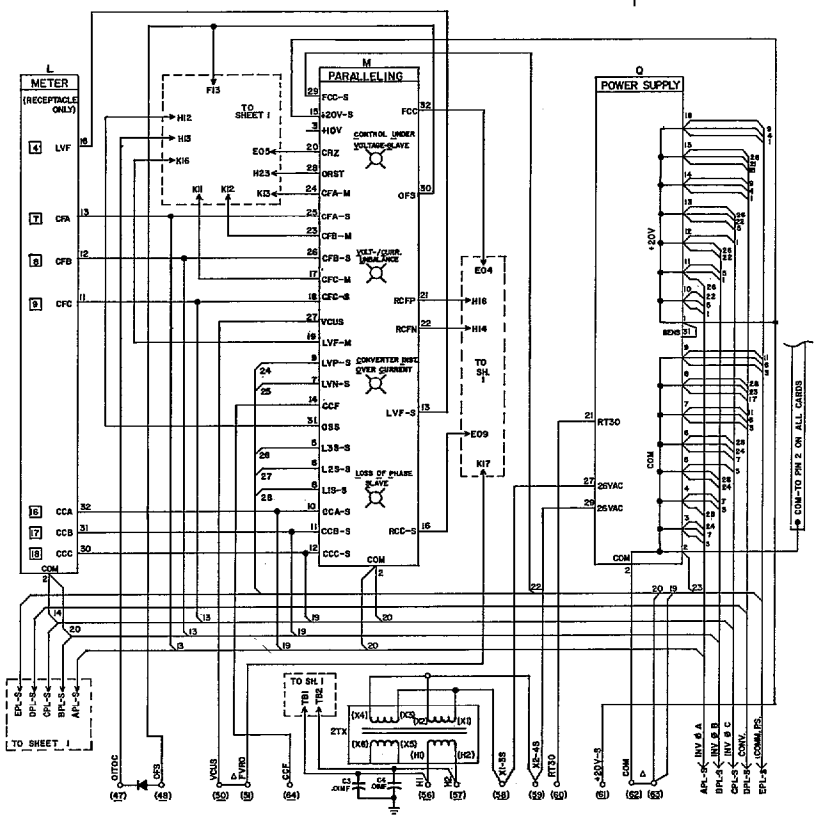
M. Gatto OCT 25,1977 — APPROVED — SPEED VARIATOR — ERIE, PA. — 36D870002AA

FP-813-F (8-78) PRINTED IN U.S.A.

GENERAL ELECTRIC — 36D870002AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE ELEMENTARY DIAGRAM — DRIVER FOR PARALLELED INVERTERS
FIRST MADE FOR AF-400 — CONT. ON 36D870002AA (A SIZE SH 3-10)



DRIVER NOMENCLATURE (CONT.)

—M SUFFIX — DENOTES ASSOCIATED WITH MASTER UNIT
—S SUFFIX — DENOTES ASSOCIATED WITH SLAVE UNIT

CCF- CONVERTER CURRENT FEEDBACK SIGNAL.
CDC- CONVERTER INSTANTANEOUS OVERCURRENT TRIP SIGNAL
OFS- FAULT SHUTDOWN SIGNAL
LOP-3- LOSS OF PHASE OR AC UNDERVOLTAGE TRIP SIGNAL
VCU- VOLTAGE OR CURRENT UNBALANCE TRIP SIGNAL
VCUS- VOLTAGE OR CURRENT UNBALANCE SENSITIVITY

No revisions are to be made to this drawing without the specific approval of the Drive/Variator Engineering Section of the Speed Variator Department.

REVISIONS | PER SVRT24428

PRINTS TO
5 (C) VA(D)
5B(BK)
5D(BK)
5E (D)
5G(BK)
5F (D)

MFG PLD Lindsay 10-12-77 — APPROVALS — SPEED VARIATOR — ERIE, PA. — 36D870002AA

REVISIONS

IN THESE NOTES, A "HIGH" LOGIC STATE REFERS TO +18 TO +20 VOLTS AND A "LOW" LOGIC STATE REFERS TO 0 TO +1 VOLT.

A. DRIVER INPUTS

1. ODMF - IF TB10 IS TAKEN LOW WHEN THE INVERTER IS RUNNING, THE INVERTER WILL DECELERATE TO ITS MINIMUM FREQUENCY AT A FAST RATE, AND WILL REMAIN THERE AS LONG AS THE ODMF INPUT IS IN THE LOW STATE. WHEN ODMF GOES HIGH, THE INVERTER WILL RETURN TO THE REFERENCE LEVEL FREQUENCY AT A NORMAL ACCELERATION RATE.

2. OSYNC - IF TB11 IS TAKEN LOW WHEN THE INVERTER IS RUNNING, THE INVERTER WILL ACCELERATE AT A FAST RATE TO THE A.C. SUPPLY LINE FREQUENCY, AND SYNCHRONIZE TO THAT FREQUENCY. THE INVERTER WILL REMAIN SYNCHRONIZED TO THE LINE FREQUENCY AS LONG AS THE OSYNC INPUT REMAINS IN THE LOW STATE, AND WILL RETURN TO THE REFERENCE LEVEL FREQUENCY AT A NORMAL RATE WHEN OSYNC GOES HIGH.

3. OSTOP - IF TB12 IS TAKEN LOW WHEN THE INVERTER IS RUNNING, THE INVERTER WILL DECELERATE TO ITS MINIMUM FREQUENCY AT A NORMAL RATE AND THEN STOP OPERATING. THE INVERTER WILL REMAIN OFF AS LONG AS THE OSTOP INPUT IS LOW IRRESPECTIVE OF ANY OTHER INPUT SIGNALS. THE ABOVE STOPPING PROCEDURE WILL ALSO OCCUR WHEN OSTOP MOMENTARILY GOES LOW IF THE OSTART INPUT IS IN THE HIGH STATE.

4. OSTART - IF TB13 IS MOMENTARILY TAKEN LOW WHEN THE INVERTER IS OFF AND THE OSTOP (TB12) INPUT IS IN THE HIGH STATE, THE INVERTER WILL START OPERATING AT ITS MINIMUM FREQUENCY, AND WILL ACCELERATE AT A NORMAL RATE TO THE REFERENCE LEVEL FREQUENCY.

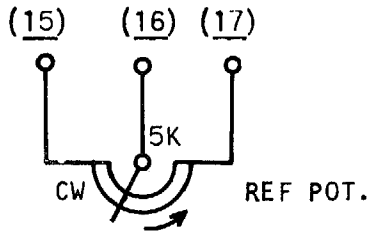
5. IXFR - IF TB14 IS HELD LOW, A FAULT SHUTDOWN CANNOT BE RESET UNTIL THE EXTERNAL FAULT RESET (TB14) INPUT IS MOMENTARILY TAKEN HIGH. WHEN TB14 IS TAKEN HIGH, OR IS LEFT UNCONNECTED, A FAULT SHUTDOWN WILL BE RESET WHEN THE OSTOP INPUT IS LOW AND THE LOGIC SIGNALS ORI AND 1MR ARE IN THEIR HIGH STATES.

6. REF, RMIN, RMAX - THE REFERENCE VOLTAGE LEVEL AT TB16 DETERMINES THE INVERTER FREQUENCY AND VOLTAGE WHEN THE INVERTER IS OPERATING NORMALLY (ODMF AND OSYNC INPUTS BOTH HIGH). IF A REFERENCE POTENTIOMETER IS USED, IT SHOULD BE 5000 OHMS IN VALUE AND HAVE ITS MINIMUM END CONNECTED TO RMIN (TB15) AT ZERO VOLTS, AND ITS MAXIMUM END CONNECTED TO RMAX (TB17), AS SHOWN IN THE SKETCH BELOW, SO THAT THE MAXIMUM REFERENCE VOLTAGE IS +15 VOLTS. IF SOME OTHER OHMIC VALUE POTENTIOMETER IS USED, ITS MAXIMUM END SHOULD BE CONNECTED IN SERIES WITH A SUITABLE VALUE RESISTOR TO +20V (TB7) SO THAT THE MAXIMUM REFERENCE VOLTAGE IS +15 VOLTS. IF SOME OTHER VOLTAGE SOURCE IS USED AS A REFERENCE, IT SHOULD BE CONNECTED BEWEEN TB16 AND TB15 (TB16 POSITIVE) AND SHOULD BE 15 VOLTS FOR MAXIMUM REFERENCE.

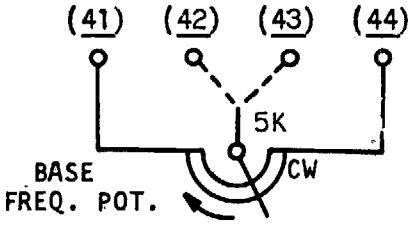
JA(CD)
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6. (CONT.)



7. BFP, BFI, BFD - IF AN EXTERNAL BASE FREQUENCY ADJUSTING POTENTIOMETER IS USED, IT SHOULD BE 5000 OHMS IN VALUE AND SHOULD BE CONNECTED FROM BFP (TB41) TO +10V (TB44) AS SHOWN IN THE SKETCH BELOW. ITS WIPER ARM SHOULD BE CONNECTED EITHER TO BFI (TB42) TO INCREASE THE THE BASE FREQUENCY OR TO BFD (TB43) TO DECREASE THE BASE FREQUENCY. THIS WILL PROVIDE EXTERNAL ADJUSTMENT OF THE BASE FREQUENCY BY AS MUCH AS 50% ABOVE OR BELOW THE INTERNALLY SET BASE FREQUENCY, WITHIN THE MAXIMUM FREQUENCY RATING OF THE DRIVE. THE JUMPER BETWEEN TB43 AND TB44 SHOULD BE REMOVED ONLY IF THE EXTERNAL POTENTIOMETER WIPER IS CONNECTED TO TB43.



B. DRIVER LINE SYNCH AND FEEDBACKS

NOTE: THE FOLLOWING DESCRIPTION IS FOR MASTER UNIT FEEDBACKS, SLAVE UNIT FEEDBACKS ARE USED ONLY FOR FAULT DETECTION.

1. L1S, L2S, L3S - TO PROVIDE FOR PHASE CONTROL LINE SYNCHRONIZATION OF THE CONVERTER; DRIVER PLUG EPL IS CONNECTED TO L1, L2 AND L3 PHASES RESPECTIVELY OF THE A.C. SUPPLY THROUGH SUITABLE SERIES RESISTORS. THESE RESISTORS ARE PLACED IN THE CONNECTING WIRE HARNESS AND ARE 1.5 MEGOHMS FOR A 230 VOLT SUPPLY AND 3.0 MEGOHMS FOR A 460 VOLT SUPPLY.

2. CVP, CVN - TO PROVIDE CONVERTER VOLTAGE FEEDBACK, DRIVER PLUG EPL IS CONNECTED TO THE CONVERTER MODULE OUTPUT, WITH CVP POSITIVE, THROUGH SUITABLE SERIES RESISTORS. THESE RESISTORS ARE PLACED IN THE CONNECTING WIRE HARNESS AND ARE 1.5 MEGOHMS FOR A 230 VOLT CONVERTER AND 3.0 MEGOHMS FOR A 460 VOLT CONVERTER.

3. LVP, LVN - TO PROVIDE D.C. LINK VOLTAGE FEEDBACK, DRIVER PLUG EPL IS CONNECTED TO THE INVERTER D.C. LINK, WITH LVP POSITIVE, THROUGH SUITABLE SERIES RESISTORS. THESE RESISTORS ARE PLACED IN THE

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CONNECTING WIRE HARNESS AND ARE 1.5 MEGOHMS FOR A 300 VOLT D.C. LINK AND 3.0 MEGOHMS FOR A 600 VOLT D.C. LINK.

4. CCF - IF CONVERTER OVERCURRENT PROTECTION IS INCLUDED, TB64 AND TB63 ARE CONNECTED TO THE OUTPUTS OF THE CONVERTER CURRENT FEEDBACK CARDS (IF PROVIDED), WITH TB64 POSITIVE.

C. DRIVER JUMPER SELECTIONS

1. BFR - TB35 IS THE MEANS TO CHANGE THE INVERTER BASE FREQUENCY RANGE AS FOLLOWS:

37.5 TO 75 HZ - JUMPER TB35 TO TB34 (COM)
75 TO 150 HZ - LEAVE TB35 OPEN.

BASE FREQUENCY IS THE FREQUENCY AT WHICH THE INVERTER REACHES FULL VOLTAGE AND IS ADJUSTED BY THE BASE FREQ. POTENTIOMETER WITHIN EITHER OF THE ABOVE TWO RANGES. ALTHOUGH THIS DIAGRAM SHOWS BFR JUMPERED TO COM, CONSULT THE DRIVE ELEMENTARY DIAGRAM TO DETERMINE IF THIS JUMPER SHOULD BE CONNECTED OR NOT.
CAUTION: IMPROPER JUMPER CONNECTION MAY RESULT IN DRIVE MALFUNCTION OR DAMAGE.

2. FT75, FT110, FT165, FT275 - TB30, TB31, TB32 AND TB33 ARE THE MEANS TO SELECT AN UPPER FREQUENCY LIMIT AT WHICH THE DRIVE WILL TRIP AND SHUT DOWN, TO PREVENT MOTOR OVERSPEED. THE FREQUENCY TRIP LEVELS ARE SELECTED BY JUMPERS AS FOLLOWS:

75 HZ TRIP - NO JUMPER
110 HZ TRIP - JUMPER TB30 TO TB31
165 HZ TRIP - JUMPER TB30 TO TB32
275 HZ TRIP - JUMPER TB30 TO TB33

CONSULT THE DRIVE ELEMENTARY DIAGRAM FOR THE PROPER JUMPER PLACEMENT.
WARNING: IMPROPER JUMPER PLACEMENT MAY RESULT IN DRIVE MALFUNCTION OR PRESENT AN EQUIPMENT OR PERSONNEL HAZARD DUE TO MOTOR OVERSPEED.

D. CARD JUMPER SELECTIONS

1. 460V JUMPER ON INVERTER CARD (193X376AA G01) - THIS JUMPER SHOULD BE PRESENT ON ALL 460 VOLT OUTPUT INVERTER DRIVES FOR PROPER OPERATION. THIS JUMPER SHOULD BE REMOVED ON ALL 230 VOLT OUTPUT INVERTER DRIVES FOR PROPER OPERATION.
CAUTION: INCORRECT JUMPER CONNECTION WILL RESULT IN MALFUNCTION AND POSSIBLE DAMAGE TO THE INVERTER.

2. 60 HZ JUMPER ON CONVERTER CARD (193X377AA G01) - THIS JUMPER SHOULD BE PRESENT ON ALL DRIVES SUPPLIED FROM 60HZ A.C. POWER FOR PROPER SYNCHRONIZING OF THE INVERTER TO THE A.C. SUPPLY (WHEN THIS IS DONE). FOR DRIVES SUPPLIED FROM 50HZ A.C. POWER, THIS JUMPER SHOULD BE REMOVED.

REVISIONS

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E. ADJUSTMENTS

ALL DRIVER ADJUSTMENTS ARE MADE BY POTENTIOMETERS LOCATED ON THE REGULATOR CARD (193X378AA G01).

1. CLIM - THE CURRENT LIMIT POTENTIOMETER ADJUSTS THE CURRENT LIMIT BETWEEN 50% AND 150% OF RATED DRIVE OUTPUT CURRENT. IT OPERATES BY REDUCING INVERTER FREQUENCY AND VOLTAGE IF THE SET CURRENT LIMIT IS EXCEEDED.

2. ATIM, DTIM - THE ACCELERATION TIME AND DECELERATION TIME POTENTIOMETERS INDEPENDENTLY ADJUST THE DRIVE ACCELERATION AND DECELERATION RATES RESPECTIVELY. FOR A 15 VOLT STEP CHANGE IN DRIVER REFERENCE, THE DRIVE WILL ACCELERATE OR DECELERATE BETWEEN ZERO AND RATED OUTPUT ON LINEAR RAMPS WHICH ARE ADJUSTABLE FROM 2.5 TO 25 SECONDS EACH.

3. VB - THE VOLTAGE BOOST POTENTIOMETER ADJUSTS THE FIXED AMOUNT OF VOLTAGE WHICH IS ADDED TO THE INVERTER IRRESPECTIVE OF FREQUENCY. THE AMOUNT OF VOLTAGE BOOST CAN BE ADJUSTED FROM ZERO TO APPROXIMATELY 7% OF RATED VOLTAGE. BELOW THE MINIMUM REFERENCE LEVEL, THE VOLTAGE BOOST IS SWITCHED OFF.

4. CLST - THE CURRENT LIMIT STABILITY POTENTIOMETER IS ADJUSTED DEPENDING ON THE AMOUNT OF DRIVE INERTIA, TO STABILIZE THE CURRENT LIMIT FUNCTION. IT CONTROLS THE RATE AT WHICH THE CURRENT LIMIT CAN REDUCE THE MOTOR SPEED ON OVERLOAD.

5. V/HZ - THE VOLTS/HERTZ POTENTIOMETER PROVIDES A VERNIER ADJUSTMENT OF THE VOLTS PER HZ OF THE INVERTER WITHIN +15%, -5% OF NOMINAL.

6. MINF - THE MINIMUM FREQUENCY POTENTIOMETER ADJUSTS THE INVERTER MINIMUM FREQUENCY FROM 3% TO 12% OF SET BASE FREQUENCY. FOR REFERENCE LEVELS BELOW THE MINIMUM FREQUENCY LEVEL, ONLY THE INVERTER VOLTAGE IS DECREASED.

7. BF - THE BASE FREQUENCY POTENTIOMETER ADJUSTS THE INVERTER BASE FREQUENCY OVER A MINIMUM 2 TO 1 RANGE WITHIN EITHER OF THE TWO BASE FREQUENCY RANGES, 37.5 TO 75 HZ OR 75 TO 150 HZ, SELECTED BY THE BFR JUMPER (SEE NOTE C1).

AN EXTERNAL BASE FREQUENCY ADJUSTMENT MAY BE ADDED TO MODIFY THE INTERNAL SETTING BY AS MUCH AS PLUS OR MINUS 50%, AS EXPLAINED IN NOTE A7.

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PRINTS TO

8. VLIM (INITIAL SET-UP ADJUST) - THE VOLTAGE LIMIT POTENTIOMETER IS USED TO PREVENT SATURATION OF THE CONVERTER BY LIMITING THE REFERENCE TO THE VOLTAGE REGULATOR. IT IS ADJUSTED TO LIMIT THE MAXIMUM INVERTER A.C. OUTPUT VOLTAGE TO BE SLIGHTLY LESS THAN THE A.C. SUPPLY VOLTAGE.

CAUTION: MISADJUSTMENT OF THIS POTENTIOMETER CAN CAUSE EXCESSIVE CURRENTS DURING MOTOR TRANSFER BETWEEN LINE AND INVERTER.

F. INDICATING LIGHTS

1. ITOC (ON SYSTEM CARD) - THE INVERSE TIME OVERCURRENT TRIP LIGHT INDICATES AN INVERTER OVERCURRENT SHUTDOWN. THE DURATION OF THE OVERCURRENT CONDITION BEFORE TRIPPING IS FROM 1 MILLISECOND AT OVER 200% OF RATED CURRENT TO 1 MINUTE AT THE SET CURRENT LIMIT LEVEL.

2. IF (ON SYSTEM CARD) - THE INVERTER FREQUENCY LIGHT INDICATES THE OPERATING FREQUENCY OF THE INVERTER PHASE A CONTROL WHETHER THE INVERTER IS ACTUALLY RUNNING OR NOT. IT IS USEFUL IN DETERMINING THE MINIMUM FREQUENCY AT STANDBY.

3. SYNC (ON SYSTEM CARD) - THE INVERTER SYNCHRONIZED LIGHT INDICATES WHEN THE INVERTER IS FREQUENCY AND PHASE SYCHRONIZED TO THE A.C. LINE.

4. LOV (ON CONVERTER CARD) - THE D.C. LINK OVERVOLTAGE TRIP LIGHT INDICATES AN OVERVOLTAGE SHUTDOWN DUE TO REGENERATIVE "PUMP UP" OF THE D.C. LINK VOLTAGE.

5. PS/LOP (ON CONVERTER CARD) - THE WRONG PHASE SEQUENCE/LOSS OF PHASE LIGHT WILL INDICATE AN INCORRECT PHASE SEQUENCE OF THE A.C. SUPPLY LINE CONNECTION, OR A LOSS OF ONE OR MORE OF THE A.C. SUPPLY LINE VOLTAGES, TO THE MASTER UNIT.

6. 1FT (ON INVERTER CARD) - THE INVERTER FAULT TRIP LIGHT INDICATES AN INVERTER "SHOOT-THROUGH" FAULT SHUTDOWN DUE TO INVERTER MISOPERATION IF IT IS THE ONLY LIGHT INDICATION. HOWEVER, OPERATION OF THE LOV, COC AND CUV TRIP CIRCUITS WILL PRODUCE A "SHOOT-THROUGH" SHUTDOWN, SO THE 1FT LIGHT WILL ALSO INDICATE WHENEVER EITHER THE LOV, COC OR CUV LIGHTS INDICATE.

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PRINTS TO

7. IOF (ON INVERTER CARD) - THE INVERTER OVERFREQUENCY TRIP LIGHT INDICATES AN OVERFREQUENCY SHUTDOWN AS DETERMINED BY THE SELECTION OF THE FREQUENCY TRIP LEVEL JUMPERS, AS EXPLAINED IN NOTE C2.

8. COC (ON INVERTER CARD) - THE COMMUTATION OVERCURRENT TRIP LIGHT INDICATES AN OVERCURRENT SHUTDOWN DUE TO AN OVERVOLTAGE IN ANY ONE OF THE THREE INVERTER COMMUTATION CIRCUITS. THIS SHUTDOWN NORMALLY OCCURS FROM EXCESSIVE MOTOR CURRENT PEAKS AT OR NEAR BASE FREQUENCY, BUT CAN BE DUE TO COMMUTATION CIRCUIT MISOPERATION AT LOWER FREQUENCIES AND VOLTAGES.

9. CUV (ON INVERTER CARD) - THE CONTROL UNDERVOLTAGE TRIP LIGHT INDICATES A SHUTDOWN DUE TO LOW CONTROL VOLTAGE OR LOW FIRING SIGNAL VOLTAGE.

10. CUV-S (ON PARALLELING CARD) - THIS CONTROL UNDERVOLTAGE TRIP LIGHT INDICATES A SHUTDOWN DUE TO LOW CONTROL VOLTAGE OF THE ADDITIONAL POWER SUPPLY FOR SCR FIRING IN THE SLAVE UNIT.

11. LOP-S (ON PARALLELING CARD) - THIS LOSS OF PHASE TRIP LIGHT INDICATES AN UNDERVOLTAGE OR A LOSS OF ONE OR MORE PHASES OF THE A.C. SUPPLY TO THE SLAVE UNIT.

12. VCU (ON PARALLELING CARD) - THE VOLTAGE OR CURRENT UNBALANCE TRIP LIGHT INDICATES A SHUTDOWN DUE TO UNBALANCED VOLTAGES OR CURRENTS BETWEEN MASTER AND SLAVE UNITS.

13. CIOC (ON PARALLELING CARD) - THE CONVERTER INSTANTANEOUS OVERCURRENT TRIP LIGHT INDICATES A CONVERTER OVERCURRENT SHUTDOWN WHEN THIS CURRENT EXCEEDS 200% OF RATED. THIS IS OPERATIONAL ONLY IF THE OPTIONAL CONVERTER CURRENT FEEDBACK PROTECTIVE FUNCTION IS FURNISHED.

G. DRIVER READOUTS

THESE SIGNAL READOUTS, EXCEPT FOR FVRO, ARE DESIGNED TO DRIVE CARD MOUNTED RELAYS CONNECTED FROM +20V TO THE READOUT TERMINAL. THEY ARE EACH CAPABLE OF "SINKING" 15 MA IN THEIR LOW STATES.

1. OSRO - IF TB18 GOES LOW, IT SIGNALS THAT THE INVERTER IS FREQUENCY AND PHASE SYNCHRONIZED TO THE A.C. LINE.

2. 1FTRO - IF TB19 GOES HIGH, IT SIGNALS A FAULT TRIP SHUTDOWN OF THE DRIVE. THE CAUSE OF THE SHUTDOWN SHOULD BE SHOWN BY THE CARD INDICATING LIGHTS AS EXPLAINED IN NOTES F1 THROUGH F13.

3. ORRO - IF TB20 IS LOW, IT SIGNALS THAT THE INVERTER IS RUNNING. IF TB20 IS HIGH, IT SIGNALS THAT THE INVERTER IS STOPPED.

4. OMVFRO - IF TB21 IS LOW, IT SIGNALS THAT THE INVERTER IS AT ITS MINIMUM VOLTAGE AND FREQUENCY. IF TB21 IS HIGH IT SIGNALS THAT THE INVERTER IS ABOVE ITS MINIMUM VOLTAGE AND FREQUENCY.

REVISIONS

JA(CD)
5TS(50)
5QC(BK)S
5QC(B)
5D(BK)
5B(3)23
PRINTS TO

5. FVRO - TB22 PROVIDES AN ANALOG VOLTAGE SIGNAL EQUIVALENT TO THE INVERTER FREQUENCY WITHIN ±1% OF MAXIMUM. THE INVERTER FREQUENCY WILL BE 5 TIMES THE TB22 VOLTAGE WHEN TB35 IS JUMPERED TO TB34 (SEE NOTE C1), AND WILL BE 10 TIMES THE TB22 VOLTAGE WHEN TB35 IS LEFT OPEN.

.H. DIAGNOSTICS

TB36 THROUGH TB40 ARE PROVIDED FOR DIAGNOSTIC CHECKOUT OF THE DRIVE USING A GOOD QUALITY OSCILLOSCOPE HAVING APPROXIMATELY A 1 MEGOHM INPUT RESISTANCE. THESE FIVE DIAGNOSTIC TERMINALS ARE CONNECTED TO THEIR CARD SIGNALS THROUGH 15K RESISTORS TO PREVENT CONTROL MISOPERATION FROM NOISE OR OTHER SIGNALS INTRODUCED AT THESE TERMINALS. HOWEVER, EXTREME CARE SHOULD STILL BE EXERCISED WHEN USING SEL1 AND SEL2 DIAGNOSTIC TERMINALS.

CAUTION: CONSULT INSTRUCTION BOOK FOR PROPER PROCEDURE IN USING SEL1 AND SEL2 FOR DIAGNOSTIC CHECKOUT SINCE DRIVE MISOPERATION AND POSSIBLE DAMAGE MAY RESULT FROM INCORRECT PROCEDURE.

1. LPAD - TB36 PROVIDES A SQUARE WAVE FREQUENCY SIGNAL WHICH IS IN PHASE WITH PHASE A (OR PHASE 1) OF THE A.C. SUPPLY LINE.

2. IPAD - TB37 PROVIDES A SQUARE WAVE FREQUENCY SIGNAL WHICH IS IN PHASE WITH PHASE A OF THE INVERTER.

3. OCPD - TB38 PROVIDES A LOW GOING COMMUTATION PULSE FREQUENCY WHICH IS 6 TIMES THE INVERTER FUNDAMENTAL FREQUENCY, AND IS USEFUL AS AN OSCILLOSCOPE TRIGGER SIGNAL.

4. SEL1, SEL2 - TB39 AND TB40 ARE CONNECTED TO TWO SELECTOR PROBES. WHICH CAN BE USED TO READ ANY CARD OUTPUT TERMINAL ON THE DRIVER BACK PLANE, BY MEANS OF AN OSCILLOSCOPE.

5. METER CARD - AN OPTIONAL METER CARD (193X381AA G01) CAN BE USED IN PLACE OF OR IN CONJUNCTION WITH AN OSCILLOSCOPE FOR DIAGNOSTIC CHECKOUT. CONSULT THE INSTRUCTION BOOK FOR DIAGNOSTIC PROCEDURES.

J. POWER SUPPLY INPUTS/OUTPUTS

1. H1, H2 - TB1, TB2, TB56 AND TB57 ARE FOR THE 115 VOLT A.C. SUPPLY TO THE TWO DRIVER CONTROL POWER TRANSFORMERS.

2. X1-3, X2-4, X1-3S, X2-4S - TB3, TB4, TB58 AND TB59 ARE FOR THE SEPARATE 28 VOLT AC OUTPUTS OF THE TWO TRANSFORMERS INTO THE TWO POWER SUPPLY CARDS.

JA(CD)
5TS(50)
5QC(BK)S
5QC(B)
5D(BK)
5B(3)23
PRINTS TO

REVISIONS

3. RT30, RT30-S - TB5 AND TB60 MAY BOTH BE CONNECTED TO A D.C. BATTERY SOURCE OF 27 TO 31 VOLTS FOR CONTROL RIDE THROUGH OF AN A.C. POWER OUTAGE.

4. PSR - TB6 MAY BE USED AS THE POWER SUPPLY REGULATING SIGNAL TO A SLAVED POWER SUPPLY.

5. +20V, +20V-S, COM - TB7, TB8, TB61, TB62 AND TB63 ARE THE SEPARATE +20 VOLT AND COMMON OUTPUTS OF THE TWO 2.5 AMP MAXIMUM RATED POWER SUPPLY CARDS.

PRINTS TO

JA(CD)

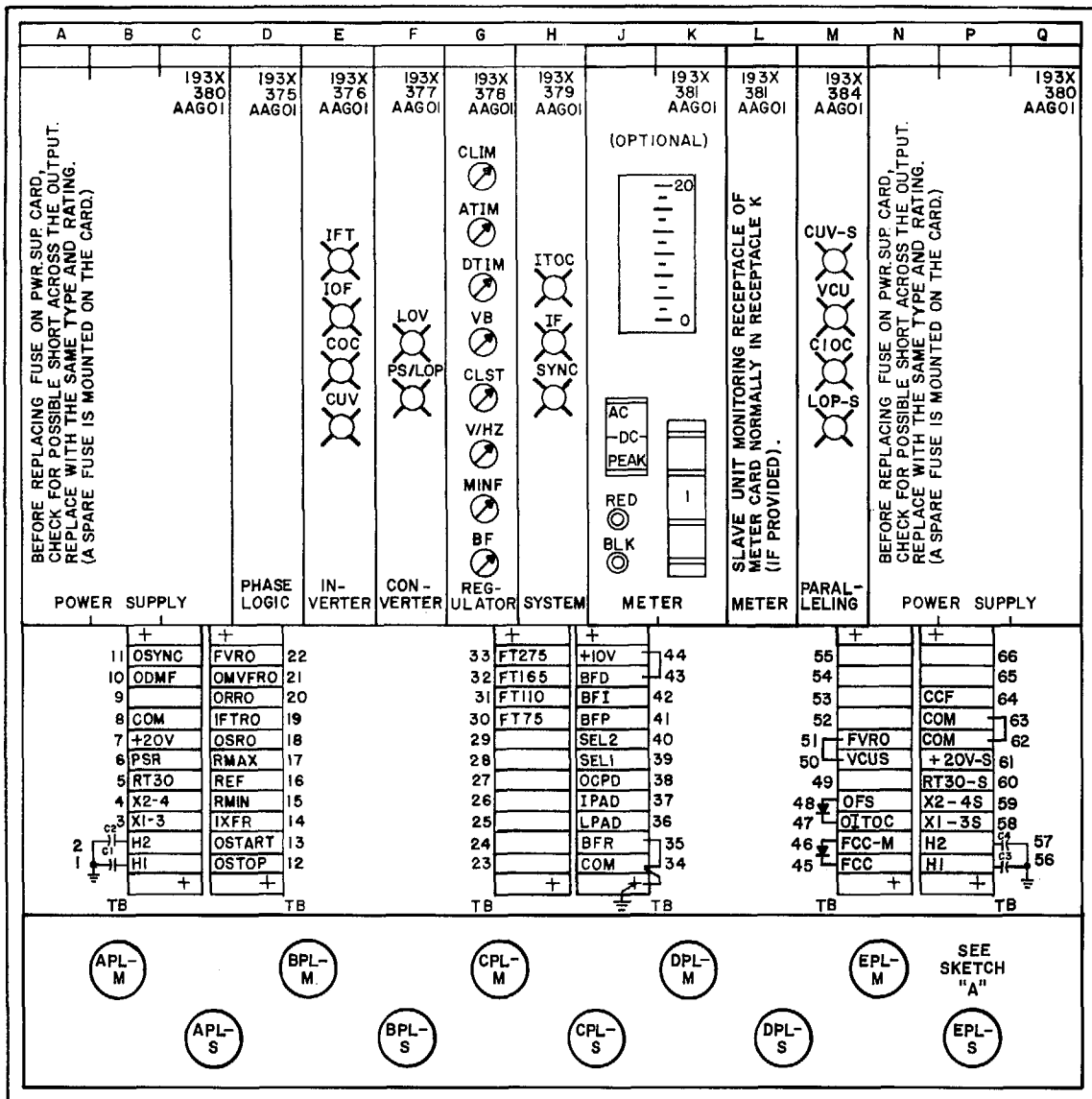
5T3(50)

50C(BK)5

50C(B)

5D(BK)

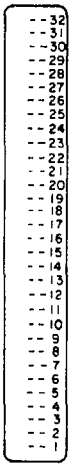
5B(3)23

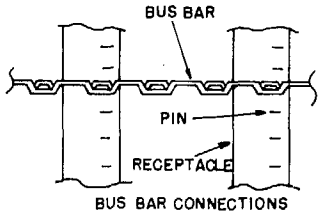


ON PRINTED CIRCUIT CARDS USED IN THIS RACK THE LETTERS "AA" AFTER BASIC CATALOG NUMBER INDICATES ORIGINAL DESIGN. SUBSEQUENT DESIGNS WITH THE SAME BASIC NUMBERS AND GROUP NUMBER WITH THE SECOND LETTER CHANGED, SUCH AS AB, AC, AD, ETC., ARE DIRECTLY INTERCHANGEABLE AND MAY BE SUPPLIED IN PLACE OF THE "AA" CARDS.

THE PRINTED CIRCUIT CARD SHOULD ALWAYS BE REMOVED WITH THE CARD EXTRACTOR WHICH IS ATTACHED ON TOP OF THE CARD RACK. SOME CARDS CONTAIN PARTS WHICH WILL BE THERMALLY HOT AFTER BEING IN OPERATION. CARE SHOULD BE EXERCISED IN HANDLING ALL CARDS AFTER REMOVAL UNTIL THESE PARTS HAVE COOLED.

FRONT VIEW OF 32 PIN RECEPTACLE AS SEEN IN RACK CLOSED POSITION.

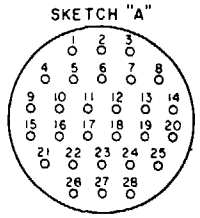




BUS BAR CONNECTIONS

FACTORY NOTES:
1. ASSEMBLE BUS BAR BEFORE WIRING. ALWAYS ASSEMBLE WITH THE BODY OF THE BUS TOWARD THE LOWER NUMBERED SIDE OF THE RECEPTACLE

SKETCH "A"



FRONT VIEW FROM PLUG SIDE

SYMBOLS

- POT ADJUSTMENT
- INDICATING LIGHT

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REV. 3	REV. 2	REV. 1 M. Gatty 5-6-85 SVB72 #28	APPROVALS	GENERAL ELECTRIC SPEED VARIATOR PROD DEPT ERIE, PA., U.S.A.	CONNECTION DIAGRAM AF-400 DRIVER
REV. 4	ISSUED O. Benson Oct 26, 1977	PRINTS TO JA(CD) 5(S)	FIRST MADE FOR REQ AF-400		
REV. 5	DRAWN M. Gatty OCT. 12, 1977	SCB(3) Z3 SCC(BE) 5 SCC(K) SCS(CD) SCC(E)	331X441AAG01	368590247AA CONT ON SH 2	SH. NO 1

SH. NO. 1
368590247AA CONT ON SH 2
REV NO 0

BUS BAR TABLE		
TABS	LOCATION	CIRCUIT
01	C D E F G H K L M	+20V
02	C D E F G H K L M Q	COM
03	F G H K L M	+10V

METAL JUMPER TABLE
TB34 TO TB MTG. SCREW
TB34 TO TB 35
TB43 TO TB44
TB50 TO TB51
TB62 TO TB63

BACK-PLANE WIRING
#24 WHITE
C01-C31
C32-E05-M20
D20-E31
D21-E16
D22-E32
D23-E19
D24-E26
D25-E27-F11-K06
D27-H24
D28-E10-F32-K07
D29-E17
D30-E18-K05
D31-H30-K09
D32-E28-H08-H09
E04-M32
E11-K32
E12-K31
E13-K30
E14-F04-H23-M28
E15-F07
E20-G18-K19
E21-F15-G22-H26-K16-M19
E29-F09
E30-F06
E09-M16
F08-G20-H27
F12-H10
F18-G24
F21-G19-K15
F25-G21-K14
F13-M30
G05-K17
G06-H31
G07-G23
G10-H04
G14-H05
G15-H07-K10
G16-H11-K08
G17-H12-M31
G26-K11-M17
G27-K12-M23
G28-K13-M24
G29-H17
G30-K18
G32-H28
G04-H16-M21
G25-H14-M22
L11-M18
L12-M26
L13-M25
L16-M13
L30-M12
L31-M11
L32-M10
M15-Q01
Q01-Q31

PLUG WIRING	
#24 WHITE	
APLM01-C10	APLS01-Q10
APLM02-D09	APLS02-D09
APLM03-C03	APLS03-Q03
APLM05-C10	APLS05-Q10
APLM06-D08	APLS06-D08
APLM07-C03	APLS07-Q03
APLM13-G28	APLS13-M25
APLM14-H02	APLS14-K02
APLM19-E11	APLS19-M10
APLM20-E02	APLS20-M02
APLM22-C10	APLS22-Q10
APLM23-D06	APLS23-D06
APLM24-C03	APLS24-Q03
APLM26-C11	APLS26-Q11
APLM27-D05	APLS27-D05
APLM28-C04	APLS28-Q04
BPLM01-C11	BPLS01-Q11
BPLM02-D17	BPLS02-D17
BPLM03-C04	BPLS03-Q04
BPLM05-C11	BPLS05-Q11
BPLM06-D18	BPLS06-D18
BPLM07-C04	BPLS07-Q04
BPLM13-G27	BPLS13-M26
BPLM14-G02	BPLS14-K02
BPLM19-E12	BPLS19-M11
BPLM20-D02	BPLS20-L02
BPLM22-C12	BPLS22-Q12
BPLM23-D11	BPLS23-D11
BPLM24-C05	BPLS24-Q05
BPLM26-C12	BPLS26-Q12
BPLM27-D10	BPLS27-D10
BPLM28-C05	BPLS28-Q05
CPLM01-C12	CPLS01-Q12
CPLM02-D15	CPLS02-D15
CPLM03-C05	CPLS03-Q05
CPLM05-C13	CPLS05-Q13
CPLM06-D16	CPLS06-D16
CPLM07-C06	CPLS07-Q06
CPLM13-G26	CPLS13-M18
CPLM14-F02	CPLS14-L02
CPLM19-E13	CPLS19-M12
CPLM20-D02	CPLS20-M02
CPLM22-C13	CPLS22-Q13
CPLM23-D13	CPLS23-D13
CPLM24-C06	CPLS24-Q06
CPLM26-C13	CPLS26-Q13
CPLM27-D12	CPLS27-D12
CPLM28-C06	CPLS28-Q06
DPLM01-C14	DPLS01-Q14
DPLM02-F27	DPLS02-F27
DPLM03-C07	DPLS03-Q07
DPLM04-C14	DPLS04-Q14
DPLM05-F26	DPLS05-F26
DPLM06-C07	DPLS06-Q07
DPLM09-C14	DPLS09-Q14
DPLM10-F29	DPLS10-F29

PLUG WIRING	
#24 WHITE	
DPLM11-C07	DPLS11-Q07
DPLM15-C15	DPLS15-Q15
DPLM16-F28	DPLS16-F28
DPLM17-C08	DPLS17-Q08
DPLM21-C15	DPLS21-Q15
DPLM22-F31	DPLS22-F31
DPLM23-C08	DPLS23-Q08
DPLM26-C15	DPLS26-Q15
DPLM27-F30	DPLS27-F30
DPLM28-C08	DPLS28-Q08
EPLM01-C16	EPLS01-Q16
EPLM02-D07	EPLS02-D07
EPLM03-C09	EPLS03-Q09
EPLM04-C16	EPLS04-Q16
EPLM05-D19	EPLS05-D19
EPLM06-C09	EPLS06-Q09
EPLM09-C16	EPLS09-Q16
EPLM10-D14	EPLS10-D14
EPLM11-C09	EPLS11-Q09
EPLM22-TB46	EPLS22-M29
EPLM23-E02	EPLS23-Q02
EPLM19-F20	EPLS19-TB63
EPLM20-F19	EPLS20-TB63
EPLM24-F16	EPLS24-M09
EPLM25-F17	EPLS25-M07
EPLM26-F24	EPLS26-M05
EPLM27-F22	EPLS27-M06
EPLM28-F23	EPLS28-M08

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

SH. NO. 2

368590247AA CONT. ON SH. 3

REV. 3	REV. 2	REV. 1	APPROVALS	GENERAL ELECTRIC	AF-400 DRIVER WIRE TABLE
REV. 4	ISSUED J. Bender Oct 26, 1977	PRINTS TO 5(S) 3A(CD)	FIRST MADE FOR REQ. AF-400	SPEED VARIATOR DEPT. ERIE, PA. U.S.A.	368590247AA CONT. ON SH. 3 / SH. NO. 2
REV. 5	MADE BY W Banks Oct. 12, 1977	58(3)23 50C(AK)5	331X441AAG01		

SV-167(5-69)

TERM. BD. WIRING	
#24 WHITE	
TB03-C27	
TB04-C29	
TB05-C21	
TB06-C25	
TB07-C01	
TB08-C02	
TB10-H32	
TB11-H25	
TB12-H21	
TB13-H20	
TB14-H22	
TB15-H02	
TB16-G30	
TB17-*	
TB18-H06	
TB19-H29	
TB20-H18	
TB21-H19	
TB22-G05	

TERM. BD. WIRING	
#24 WHITE	
TB30-E22	
TB31-E25	
TB32-E24	
TB33-E23	
TB34-G02	
TB35-G13	
TB36-*	
TB37-*	
TB38-*	
TB39-*	
TB40-*	
TB41-G09	
TB42-G11	
TB43-G12	
TB44-G03	
TB45-M32	
TB47-H13	
TB48-M30	
TB50-M27	
TB51-K17	
TB56-TB1	
TB57-TB2	
TB58-Q27	
TB59-Q29	
TB60-Q21	
TB61-Q01	
TB62-Q02	
TB64-M14	

WIRE TABLE

1TX LEADS

LEAD	CONNECTION
(BLK)H1	TB01
(BLK)H2	TB02
(BRN)X1	TB03
(BRN)X2	TB04
(GRN)X3	TB03
(GRN)X4	TB04
(YEL)X5	CUT & INSULATE
(YEL)X6	CUT & INSULATE

2TX LEADS

LEAD	CONNECTION
(BLK)H1	TB56
(BLK)H2	TB57
(BRN)X1	TB58
(BRN)X2	TB59
(GRN)X3	TB58
(GRN)X4	TB59
(YEL)X5	CUT & INSULATE
(YEL)X6	CUT & INSULATE

C1 & C2 CONNECTIONS

LEAD	CONNECTION
C1-H1	TB01
C2-H2	TB02
C1,C2-GRD	TB MTG SCREW

C3 & C4 CONNECTIONS

LEAD	CONNECTIONS
C3-H1	TB56
C4-H2	TB57
C3,C4-GRD	TB MTG. SCREW

*CONNECT RESISTOR WIRES AS INDICATED BELOW:

RESISTOR NOMEN.		1ST END CONN.	2ND END CONN.
RMAX		TB17	H01
LPAD		TB36	F12
IPAD		TB37	H09
OCPD		TB38	K19
SEL1	(BLACK)	TB39	K13(MOVEABLE END)
SEL2	(VIOLET)	TB40	K15(MOVEABLE END)

MOVEABLE JUMPER CONNECTIONS

THE FIRST WIRE CONNECTION IS THE PERMANENT ONE.
THE SECOND WIRE CONNECTION IS THE MOVEABLE ONE.

#24 RED

K28- K20

RESISTOR WIRES

SEL1 - } SEE RESISTOR WIRE TABLE
SEL2 - }

SH. NO. 3
36B590247AA CONT. ON SH. 4

REV. 3	REV. 2	REV. 1	APPROVALS	GENERAL ELECTRIC	AF-400 DRIVER WIRE TABLE
REV. 4			FIRST MADE FOR REQ. AF-400		36B590247AA CONT. ON SH. 4 / SH. NO. 3
REV. 5			331X941AAG01		

5T3(5O) 5QC(B) 5D(BK)

DRIVER

C	D	E	F	G	H	K OR L	M	Q
193X380AAG01	193X375AAG01	193X376AAG01	193X377AAG01	193X378AAG01	193X379AAG01	193X381AAG01	193X384AAG01	193X380AAG01
BEFORE REPLACING FUSE ON POWER SUPPLY CARD, CHECK FOR POSSIBLE SHORT ACROSS THE OUTPUT. REPLACE WITH THE SAME TYPE AND RATING FUSE. (A SPARE FUSE IS MOUNTED ON THE CARD.)		INVERTER FAULT TRIP INVERTER OVERFREQUENCY COMMUTATION OVERCURRENT CONTROL UNDERVOLTAGE	D.C. LINK OVERVOLTAGE WRONG PH. SEQ LOSS OF PHASE	CURRENT LIMIT ACCEL TIME DECEL. TIME VOLTAGE BOOST CURR. LIM. STAB. VOLTS / HERTZ MINIMUM FREQ. BASE FREQ.	INVERSE TIME OVERCURRENT INVERTER FREQUENCY INVERTER SYNCHRONIZED	20 0 AC DC PEAK RED BLK I	CONTROL UNDERVOLTAGE - SLAVE VOLT./CURR. UNBAL. CONVERTER INST OVERCURRENT LOSS OF PHASE - SLAVE	BEFORE REPLACING FUSE ON POWER SUPPLY CARD, CHECK FOR POSSIBLE SHORT ACROSS THE OUTPUT. REPLACE WITH THE SAME TYPE AND RATING FUSE. (A SPARE FUSE IS MOUNTED ON THE CARD.)
POWER SUPPLY	PHASE LOGIC	INVERTER	CONVERTER	REGULATOR	SYSTEM	(OPTIONAL) METER	PARALLELING	POWER SUPPLY

INPUTS	11	OSYNC	22	FVRO	33	FT275	44	+10V	
	10	ODMF	21	OMVFRO	32	FT165	43	BFD	EXT. BASE FREQUENCY POT. CONN
	9		20	ORRO	31	FT110	42	BFI	
	8	COM	19	IFTRO	30	FT75	41	BFP	
	7	+20V	18	OSRO	29		40	SEL2	
POWER SUPPLY	6	PSR	17	RMAX	28		39	SEL1	
	5	RT30	16	REF	27		38	OCPD	DIAGNOSTIC
	4	X2-4	15	RMIN	26		37	IPAD	
	3	X1-3	14	IXFR	25		36	LPAD	
115VAC INPUT	2	H2	13	OSTART	24		35	BFR	BASE FREQ. RANGE SEL.
	1	H1	12	OSTOP	23		34	COM	

READOUTS / INPUTS — OVER FREQUENCY SELECTION

PARAL FAULT DETECT.	55			66
	54			65
	53		CCF	64
	52		COM	63
	51	FVRO	COM	62
	50	VCUS	+20V-S	61
	49		RT30-S	60
	48	OFS	X2-4S	59
	47	OITOC	X1-3S	58
	46	FCC-M	H2	57
	45	FCC	H1	56

CONV. CURR. FB. — POWER SUPPLY — 115VAC INPUT

IF ONLY THE SECOND LETTER HAS CHANGED IN THE TWO LETTER GROUP OF THE CARD 193X NO., THE CARDS ARE INTERCHANGEABLE (AB REPLACES AA).

CARD REPLACEMENT INSTR

MAKE POT SETTINGS (INCL. VLIM POT ON REGUL. CARD) AND CARD JUMPERS SAME AS ON ORIGINAL CARD.

SPACE FOR CAUTION LABEL NP104X905BA148

460V JUMPER ON INVERTER CARD

460 VOLT INVERTER - JUMPER NEEDED
230 VOLT INVERTER - REMOVE JUMPER

60HZ JUMPER ON CONVERTER CARD

60 HZ AC SUPPLY - JUMPER NEEDED
50 HZ AC SUPPLY - REMOVE JUMPER

SPACE FOR CAUTION LABEL NP104X905BA067

BASE FREQUENCY RANGE SELECTION

BASE FREQ. RANGE	37.5 TO 75HZ	75 TO 150HZ
JUMPER TB35 TO	TB34	TB35

FREQUENCY TRIP SELECTION

FREQUENCY TRIP	75HZ	110HZ	165HZ	275HZ
JUMPER TB30 TO	TB30	TB31	TB32	TB33

SPACE FOR WARNING LABEL NP104X905BA068

SELECTOR SWITCH POS.	OPTIONAL METER CARD — SELECTED SIGNAL	SCALE SELECT.	METER READING (±1.0 MAX. ERROR) — OFF CONDITION	0 REF. 0 LOAD	1/2 REF. 1/2 LOAD	1 REF. 1 LOAD	SPECIAL CONDITION
1	+10V - POWER SUPPLY MIDPOINT	DC	10	10	10	10	
2	REF - REFERENCE VOLTAGE	DC	①	0	7.5	15	
3	FVRO - INVERTER FREQUENCY	DC	0.3 TO 1.5	0.3 TO 1.5	6 ②	12 ②	
4 ⑤	LVF - DC LINK VOLTAGE	DC	0 TO 1.5	0 TO 0.5	7.5 ③	15 ③	
5	CVF - CONVERTER VOLTAGE FEEDBACK	DC	9.7	10	7.7	5.5	
6	CVR - CONVERTER VOLTAGE REFERENCE	DC	10.5	8 TO 10.5	6	3.5	
7 ⑤	CFA - AC MOTOR CURRENT - PHASE A	AC(X10)	0	0	5	10	15 AT 1.5 LOAD
8 ⑤	CFB - AC MOTOR CURRENT - PHASE B	AC(X10)	0	0	5	10	15 AT 1.5 LOAD
9 ⑤	CFC - AC MOTOR CURRENT - PHASE C	AC(X10)	0	0	5	10	15 AT 1.5 LOAD
10	OCL - CURRENT LIMIT SIGNAL	DC	15 TO 17	15 TO 17	15 TO 17	15 TO 17	4 TO 8 AT CURR. LIMIT
11	IRI - INVERTER RUN SIGNAL	DC	0	17 TO 19	17 TO 19	17 TO 19	
12	OIS - INVERTER SYNCHRONIZE SIGNAL	DC	17 TO 19	17 TO 19	17 TO 19	17 TO 19	0 AT SYNCH.
13	DFS - DELAYED FIRING SUPPLY	DC	17 TO 19	17 TO 19	17 TO 19	17 TO 19	
14	IIPT - INVERTER FIRING PULSE TRAIN	PEAK/DC	17 TO 19/1	17 TO 19/1	17 TO 19/1	17 TO 19/1	
15	ICFF - INVERTER 3 PHASE LOGIC SIGNAL	PEAK/DC	17 TO 19/0	17 TO 19/0	17 TO 19/0	17 TO 19/0	
16 ⑤	CCA - INV. PEAK COMM. CURR. - PH. A	PEAK	0	9 TO 11	11 TO 13	12 TO 14	
17 ⑤	CCB - INV. PEAK COMM. CURR. - PH. B	PEAK	0	9 TO 11	11 TO 13	12 TO 14	
18 ⑤	CCC - INV. PEAK COMM. CURR. - PH. C	PEAK	0	9 TO 11	11 TO 13	12 TO 14	
19 ④	SEL - D TAB 5, 6, 7, 10, 11, 12, 13, 14 & 19	PEAK/DC	0	15 TO 18/0	15 TO 18/0	15 TO 18/0	0 AFTER FAULT
19 ④	SEL - D TAB 8, 9, 15, 16, 17 & 18	PEAK/DC	0	16 TO 19/2	16 TO 19/2	16 TO 19/2	0 AFTER FAULT
19 ④	SEL - F TAB 26, 27, 28, 29, 30 & 31	PEAK/DC	16 TO 19/1	16 TO 19/1	16 TO 19/1	16 TO 19/1	0 AFTER FAULT

① DEPENDENT ON REFERENCE SETTING.

② READINGS SHOWN ARE FOR 60HZ BASE FREQUENCY. ACTUAL FREQUENCIES ARE 5 TIMES METER READING FOR TAB35 CONN. TO TAB34 OR 10 TIMES METER READING FOR TAB35 OPEN.

③ ACTUAL DC LINK VOLTAGES ARE 20 TIMES METER READING FOR 230 VOLT DRIVES AND 40 TIMES METER READING FOR 460 VOLT DRIVES.

④ POS. 19 IS FOR DRIVER BACKPLANE SELECTOR PROBE. READINGS SHOWN ARE SCR FIRING SIGNALS.

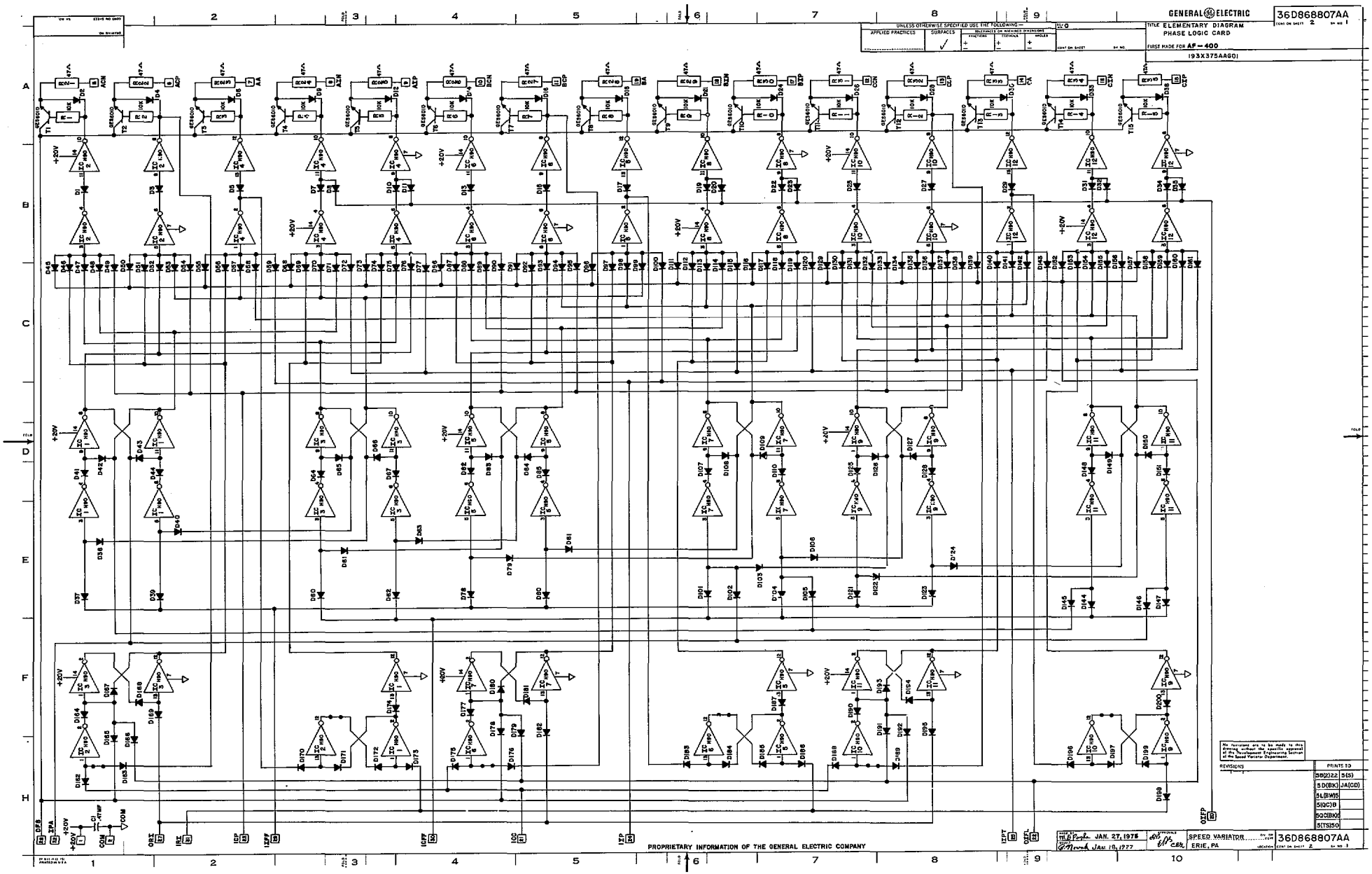
⑤ SWITCH POSITIONS FOR SLAVE UNIT READINGS WHEN METER CARD IS INSERTED IN RECEPTACLE L.

SH. 4

368590 247AA

1. SIZE: 10.75" WIDE X 16" HIGH

REV. 1 M. Cotty SV872#28	ISSUED L. Bender Oct 26, 1979	PRINTS TO S(6) A(CO) S(8)(23) S(D(BK) S(C(BR) S(TS(OR)	FIRST MADE FOR REQ. AF-400	GENERAL ELECTRIC SPEED VARIATOR PROD DEPT ERIE, PA., U.S.A.	LABEL ARTWORK AF-400 DRIVER	
REV. 2	MADE BY M. Cotty OCT. 17, 1977		331X441 AAG01		368590 247 AA (DWT OR SK.) FL	SK. NO. 4



GENERAL ELECTRIC

36D868807AA

TITLE ELEMENTARY DIAGRAM
PHASE LOGIC CARD

FIRST MADE FOR AF-400

193X375AAG01

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

SPEED VARIATOR ERIE, PA

36D868807AA

GENERAL ELECTRIC

36D868807AA

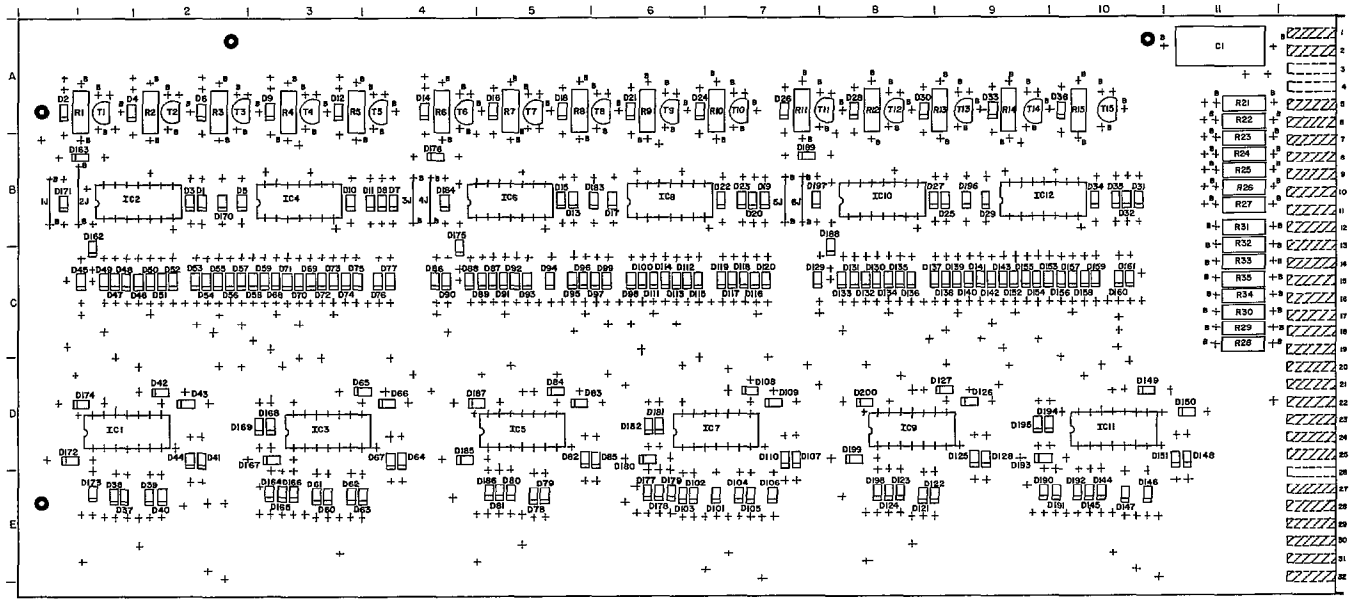
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

APPLIED PRACTICES | SURFACES

TITLE PRINTED CIRCUIT DIAGRAM
PHASE LOGIC CARD

FIRST MADE FOR AF-400

193X375AAG01



HOLE TABULATION

ALL HOLES ARE Ø32 EXCEPT THE HOLES TABULATED BELOW

LOC	DIA	QTY
A	157	4
B	040	113

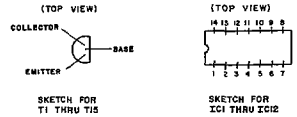
NOTES

1 INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS

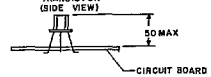
2 CROSS HATCHED TABS INDICATES TABS USED

3 CARD SIZE, H $80^{+.005}_{-.010}$ X $5\ 130^{+.005}_{-.010}$

TRANSISTOR LEAD SKETCH



COMPONENT HEIGHT SKETCH



2X	368605228AA005	368605229AA005
SCALE	REAR ETCHED BOARD DWG	FRONT ETCHED BOARD DWG

REVISIONS

PRINTS TO

5B(2)32	5B
5D(BK)	AW(BW)
5L(BW)5	JA(CD)
5(OC)B	DS(REP)
5OCB)05	
5(TS)50	n/l

E. Marsh JAN 23,1976

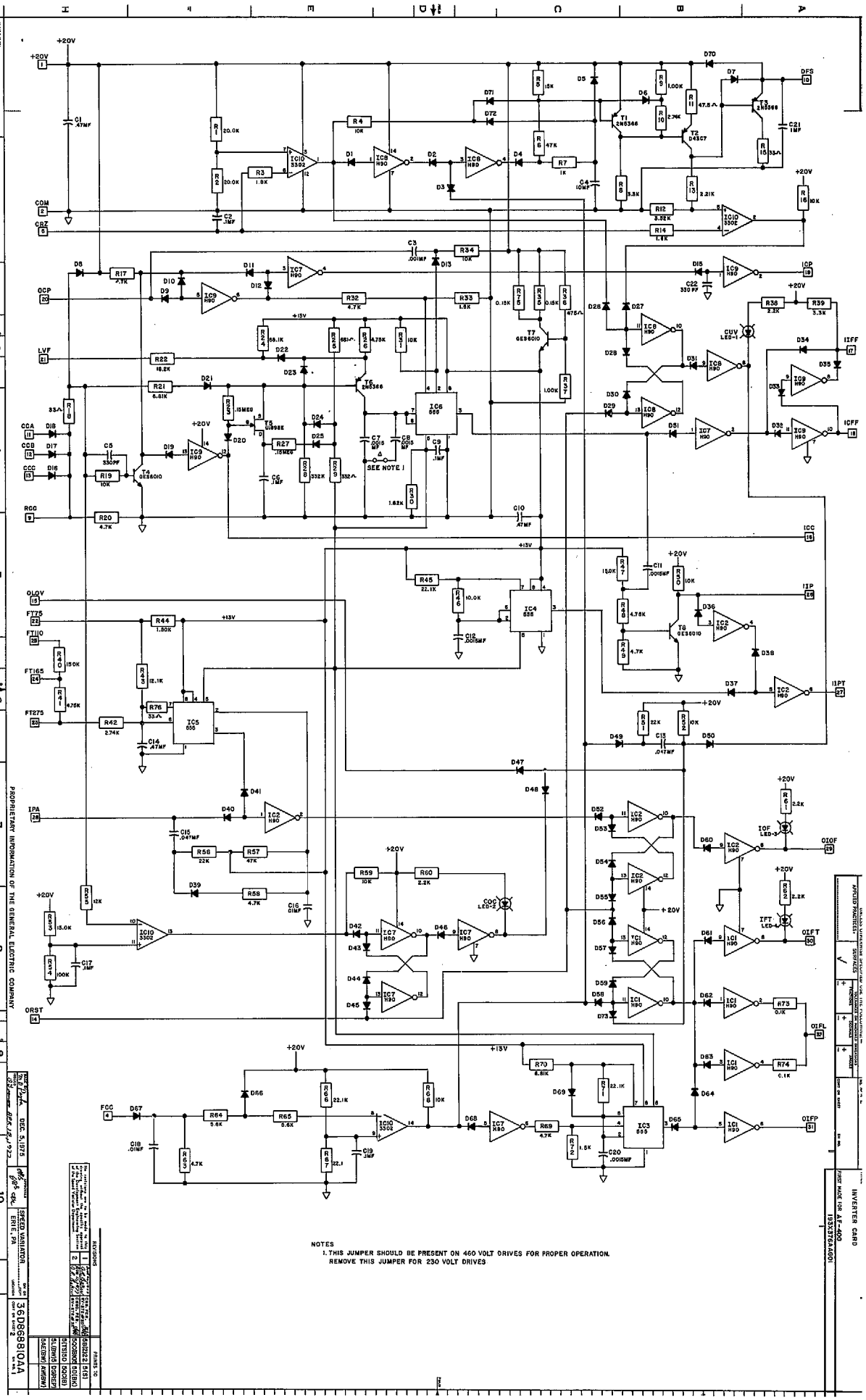
E. Marsh JAN 18,1977

SPEED : VIATOR

ERIE, PA

36D868807AA

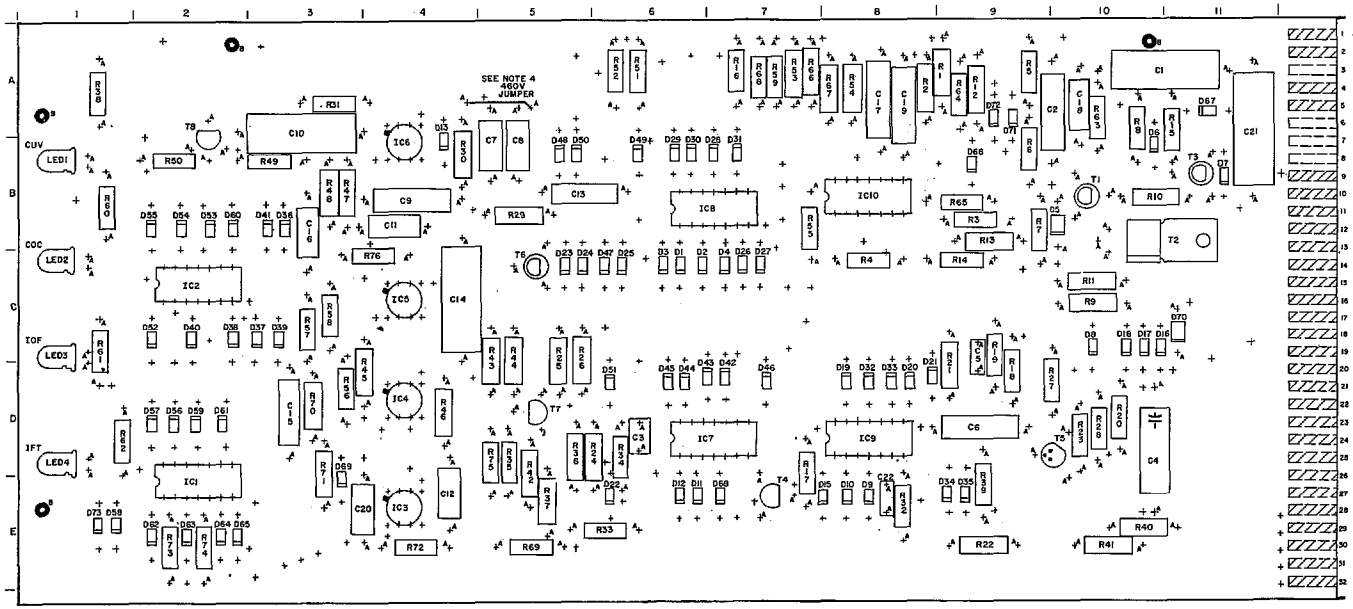
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


+20V
COM
CRZ
OCP
LVF
CCA
CCB
CCC
RGG
OLOV
FT175
FT110
FT165
FT275
IPA
ORST
FCC
+15V
DFS
ICP
ICC
IIP
IIPT
OIOF
OIFT
OIFL
OIFP
IOF LED-3
IFT LED-4
CUV LED-1
COC LED-2
IIFF
IICFF
SEE NOTE 1
T1 2N5366
T2 D43C7
T3 2N3569
T4 GES6010
T5 U1898E
T6 2N5366
T7 GES6010
T8 GES6010
IC1 H90
IC2 H90
IC3 555
IC4 556
IC5 555
IC6 556
IC7 H90
IC8 H90
IC9 H90
IC10 330E
IC10 5302
NOTES
1. THIS JUMPER SHOULD BE PRESENT ON 460 VOLT DRIVES FOR PROPER OPERATION.
REMOVE THIS JUMPER FOR 230 VOLT DRIVES
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY
SPEED VARIATOR
ERIE, PA
DEC. 5,1975
INVERTER CARD
FIRST MADE FOR AF-400
IS3X376AA401
361D868B10AA
REVISIONS
PRINTS TO

GENERAL ELECTRIC — 36D868810AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES: — SURFACES: ✓ — TOLERANCE ON MACHINED DIMENSIONS: FRACTIONS ± — DECIMALS ± — ANGLES ±

TITLE: PRINTED CIRCUIT DIAGRAM
INVERTER CARD
FIRST MADE FOR AF-400
193X376AA001



HOLE TABULATION

ALL HOLES .032 DIA. EXCEPT THE HOLES TABULATED BELOW.

A	.040	2.54
B	.157	4

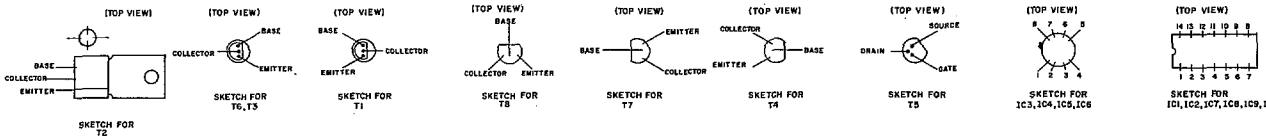
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED
3. CARD SIZE, 11.50 +.000 −.010 X 5.130 +.005 −.005
4. 460V JUMPER IS TO BE PUT ON ALL CARDS. THIS JUMPER WILL BE REMOVED FOR 230V APPLICATIONS.
5. ALL COMPONENTS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.

COMPONENT HEIGHT SKETCH
TRANSISTOR
(SIDE VIEW)



TRANSISTOR LEAD SKETCHES



1	0	2X	368605231AA005	368605232AA005
FRONT	BACK	SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.
SPACE UNIT REQUIREMENT				

REVISIONS

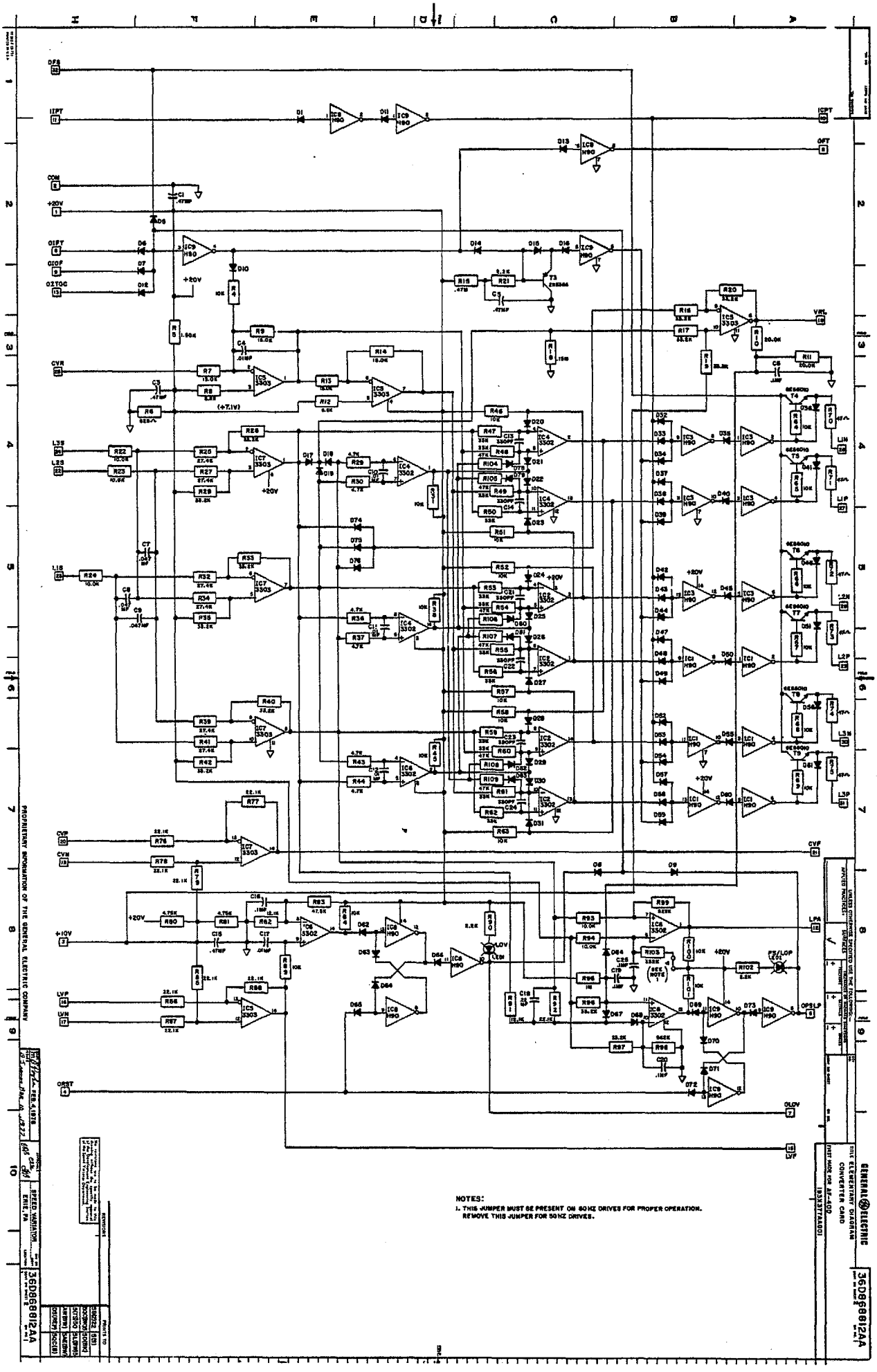
1	12 Oct 1972	CORR. PER SY-272 #38005	
2	DEC 3, 1975 12 Oct 1972	CORR. PER SY-274 #38005	

PRINTS TO

SD(81)M	SJ(BW)
SR(218W	TS(SO)
SL(218W	
AW(BW)	
SO(CO)	
JA(CP)	

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DRAWN FEB. 3, 1976 — APPROVED — SPEED VARIATOR — LOCATION ERIE, PA — 36D868810AA



NOTES:
1. THIS JUMPER MUST BE PRESENT ON 60 HZ DRIVES FOR PROPER OPERATION. REMOVE THIS JUMPER FOR 50 HZ DRIVES.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

TITLE ELEMENTARY DIAGRAM CONVERTER CARD

SPEED VARIATION ERIE, PA

36D86868012AA

GENERAL ELECTRIC

36D868812AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING: —
APPLIED PRACTICES: | SURFACES: ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE PRINTED CIRCUIT DIAGRAM
CONVERTER CARD

FIRST MADE FOR AF-400

193X377AAG01

HOLE TABULATION

ALL HOLES ARE .032 DIA. EXCEPT THE HOLES TABULATED BELOW.

LOC.	DIA.	QTY.
A	.157	4
B	.040	287

NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, 11.50 +.000 −.015 X 6.150 +.005 −.005
4. 60HZ JUMPER IS TO BE PUT ON ALL CARDS. THIS JUMPER WILL BE REMOVED FOR 50HZ APPLICATIONS.

SEE NOTE 1 & 2

SEE NOTE 4 60HZ JUMPER

LED2 PS/LOP

LED1 LOV

TRANSISTOR LEAD SKETCHES

(TOP VIEW) EMITTER COLLECTOR BASE — SKETCH FOR T3

(TOP VIEW) EMITTER COLLECTOR BASE — SKETCH FOR T1

(TOP VIEW) EMITTER BASE COLLECTOR — SKETCH FOR T4-T9

(TOP VIEW) 14 13 12 11 10 9 8 / 1 2 3 4 5 6 7 — SKETCH FOR IC1-9

EX	368605237AA005	368605238AA005
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.

REVISIONS

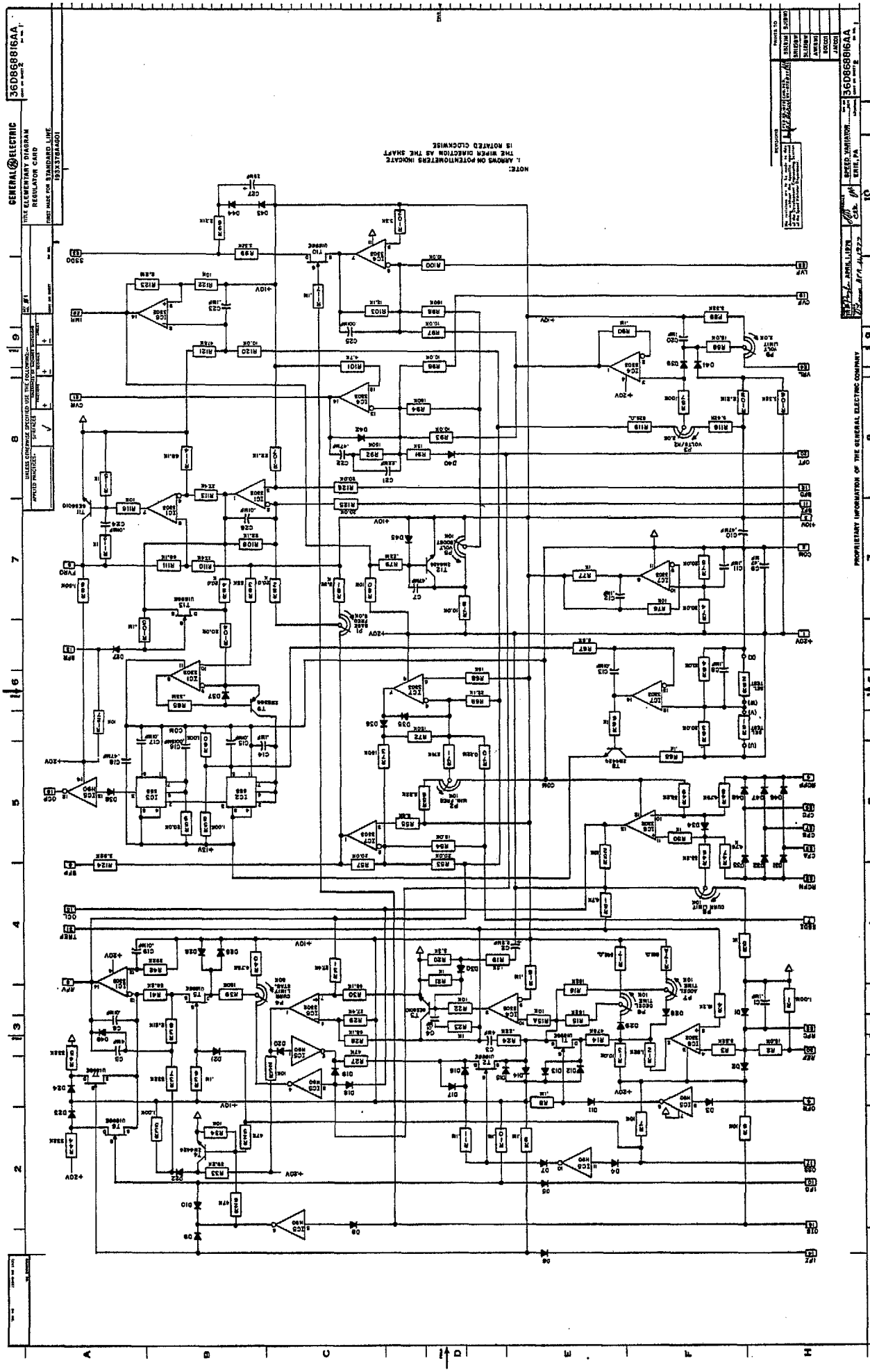
1 | MAR 7, 1978 | CHG. PER 00-777-870 | JH

PRINTS TO

SB(B)M	3J(B)M
SR(2)B)M	TR(50)
SL(2)B)M	
AW(B)M	
SD(CD)	
JA(CD)	

E Watkins JAN. 17, 1977 — Speed Variator — ERIE, PA., USA — 36D868812AA

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY


GENERAL ELECTRIC
36D86881GAA
TITLE ELEMENTARY DIAGRAM
REGULATOR CARD
FIRST MADE FOR STANDARD LINE
193X376AA001
NOTE:
1. ARROWS ON POTENTIOMETERS INDICATE THE WIPER DIRECTION AS THE SHAFT IS ROTATED CLOCKWISE
PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

360B868816AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—
APPLIED PRACTICES: SUPERSEDES ✓ | TOLERANCES ON DECIMALS: FRACTIONS ± | ANGLES ± | FINISH ± | ANGLES ±

TITLE PRINTED CIRCUIT DIAGRAM
REGULATOR CARD

FIRST MADE FOR AF-400

183X378AAG01

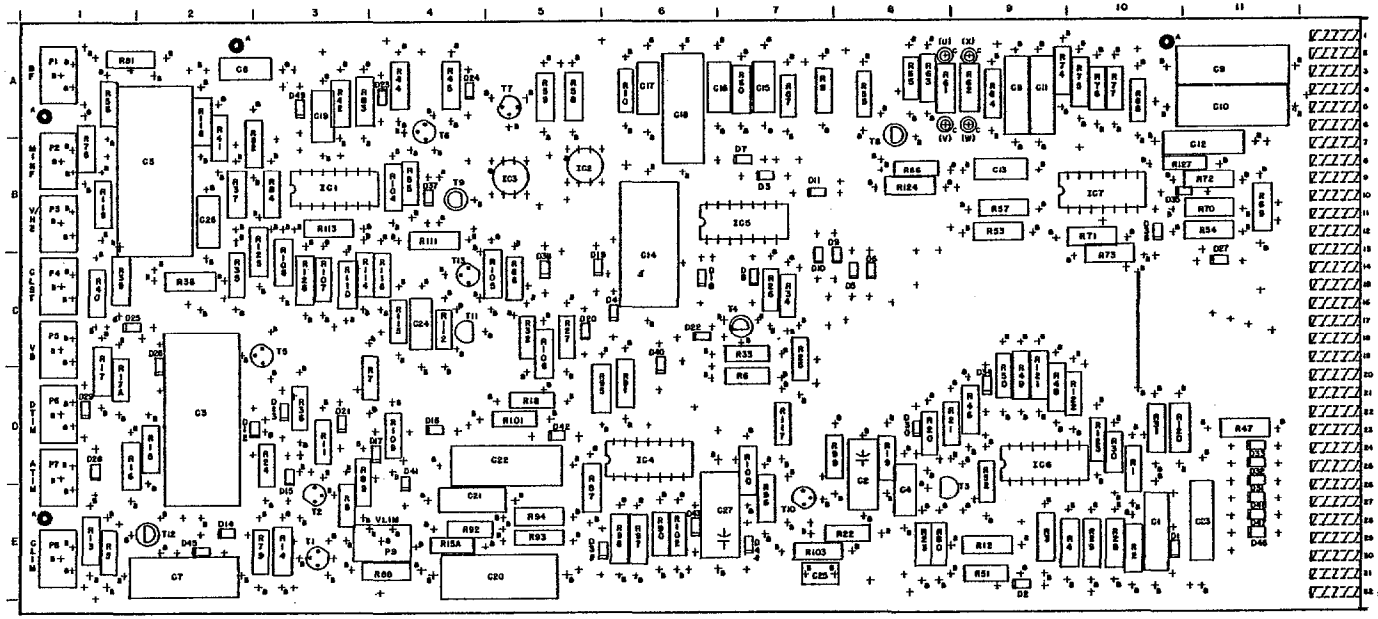
HOLE TABULATION

ALL HOLES ARE .032 EXCEPT THE HOLES TABULATED BELOW.

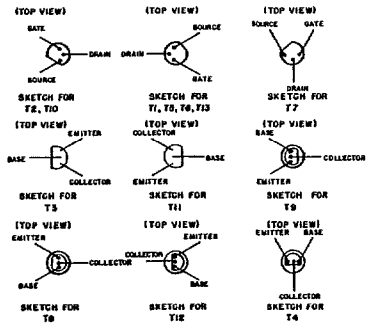
LOC.	DIA.	QTY.
A	.157	4
B	.040	374
C	.070	4

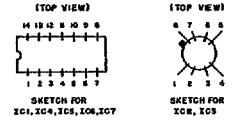
NOTES

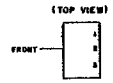
1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, 11.50 +.000 −.015 × 6.150 +.005 −.005
4. ALL COMPONENTS SHALL BE MOUNTED TO A .50 INCH MAXIMUM ABOVE THE CARD SURFACE AS SHOWN BELOW.

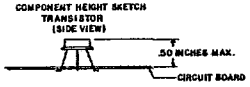


TRANSISTOR LEAD SKETCHES









1	0	2K	360B052S4AA005	360B052S5AA005
FRONT	BACK	SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.
SPACE UNIT REQUIREMENT				

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Forster Department.	1	ADD P9 (7-77) ECO 183-272 (2-77) C.E. Gaulke	
	2	ADD P10 (10-21-77) ECO 183-272 (2-21-77) C.E. Gaulke	

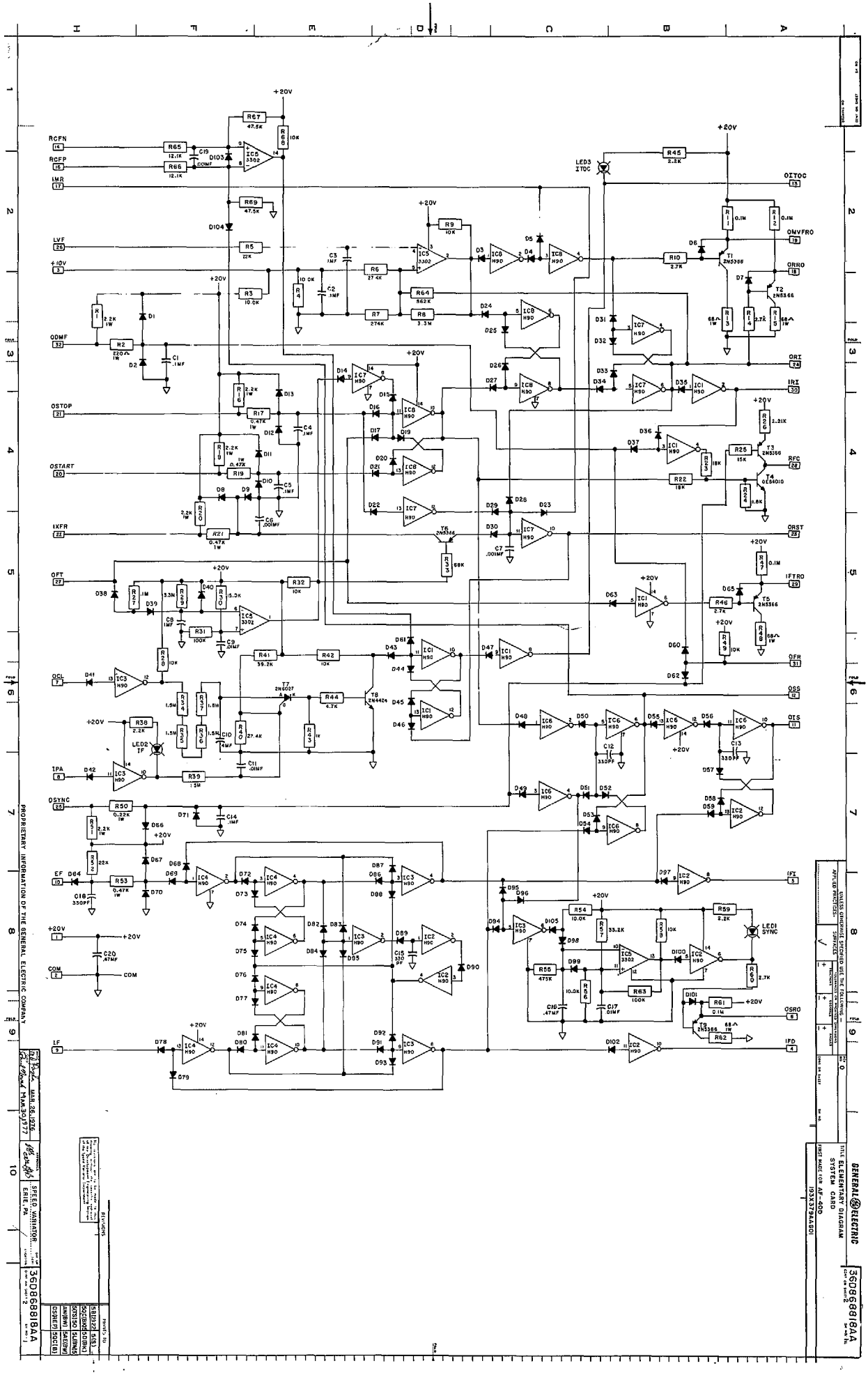
PRINTS TO

SBIBIM	SAIBIM
SRI2IBW	
SLI2IBW	
AWBIMS	
SDICOI	
JAICOI	

C.E. Gaulke FEB. 21, 1977 — speed variator — ERIE, PA., USA — 360D868816AA

ADD P10 per ECO 183-272 dated 11-22-77

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

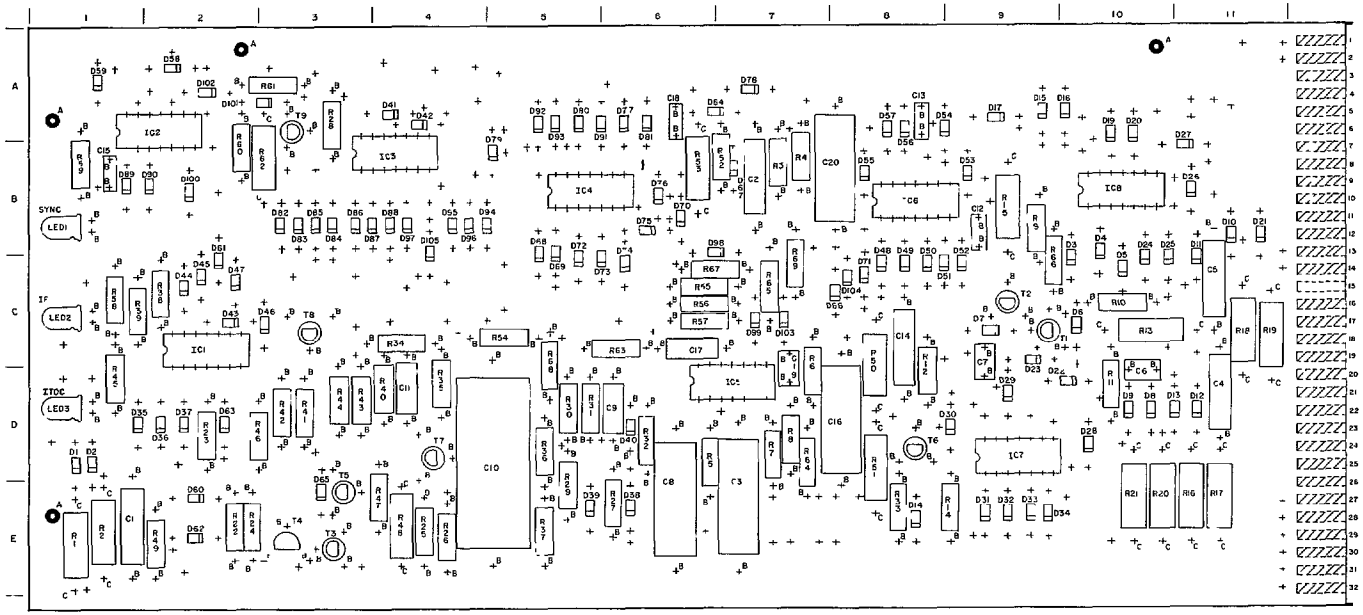


PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DRAWN	MAR.26.1976	GENERAL ELECTRIC
ENGINEER	MAR30,1977	TITLE ELEMENTARY DIAGRAM SYSTEM CARD
MFG	SPEED MONITOR ERIE, PA	FIRST MADE FOR AF-400

360B68818AA

360B68818AA



HOLE TABULATION

ALL HOLES ARE 032 EXCEPT THE HOLES TABULATED BELOW

LOC	DIA	QTY
A	157	4
B	040	183
C	052	30

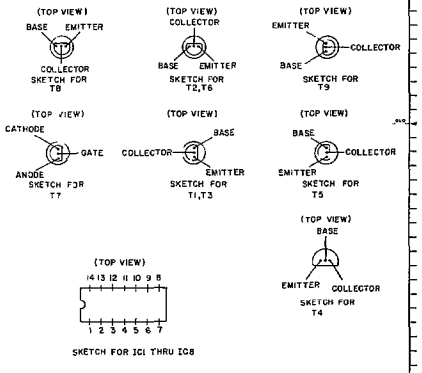
NOTES

1 INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS

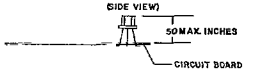
2 CROSS HATCHED TABS INDICATES TABS USED

3 CARD SIZE, 11.50 +.000 −.005 × 5.120 +.005 −.000

TRANSISTOR LEAD SKETCHES



COMPONENT HEIGHT SKETCH



	FRONT	BACK
SPACE UNIT REQUIREMENT		
2X	368605245AA005	368605246AA005
SCALE	REAR ETCHED BOARD DWG	FRONT ETCHED BOARD DWG

58(2122	5(S)
5QC(BK)	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC — 36D868818AA — FL 2

UNLESS OTHERWISE SPECIFIED USE NEXT FOLLOWING — APPLIED DIMENSIONS — TOL

TITLE PRINTED CIRCUIT DIAGRAM SYSTEM

FIRST MADE FOR AF-400

193X379AA001

Speed varrior — ERIE, PA., USA — 36D868818AA — FL 2

GENERAL ELECTRIC

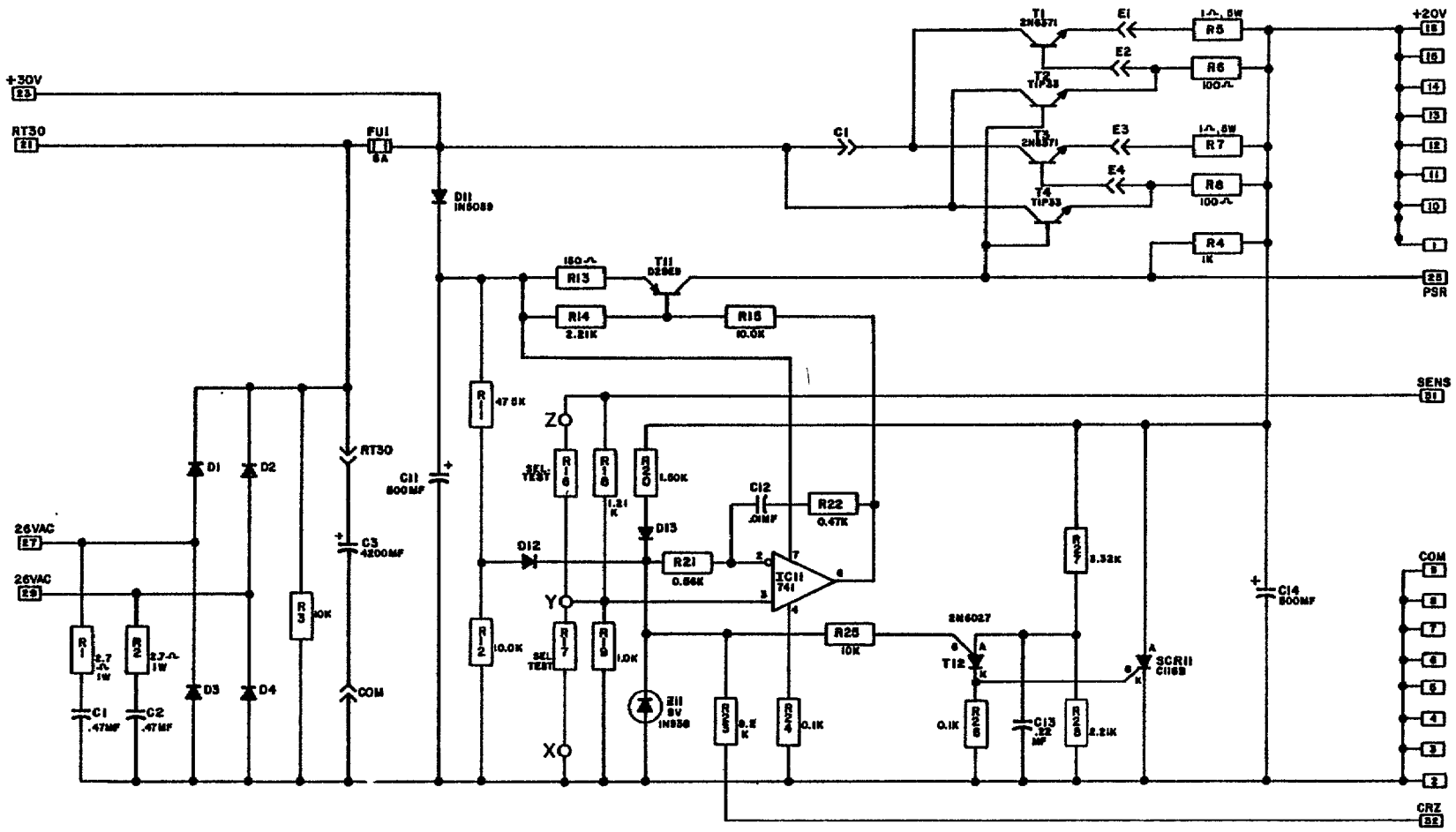
36C764164AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
POWER SUPPLY

FIRST MADE FOR AF400
193X380AAG01



REV NO.
0

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO	
5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
D3(REP)	5QC(B)

MADE BY M D Pugh MAY 10, 1976
ISSUED Paul Mark FEB. 1, 1977
APPROVALS NB Curtis

SPEED VARIATOR — DIV OR DEPT
ERIE, PA — LOCATION
36C764164AA — CONT ON SHEET — SH NO

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

FF 505 F (9 75)
PRINTED IN U S A

GENERAL ELECTRIC

36D868820AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:— APPLIED PRACTICES: SURFACES: ✓ TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± DECIMALS ± ANGLES ±

REV NO. 0

TITLE PRINTED CIRCUIT DIAGRAM POWER SUPPLY

FIRST MADE FOR AF-400

193X380AAG01

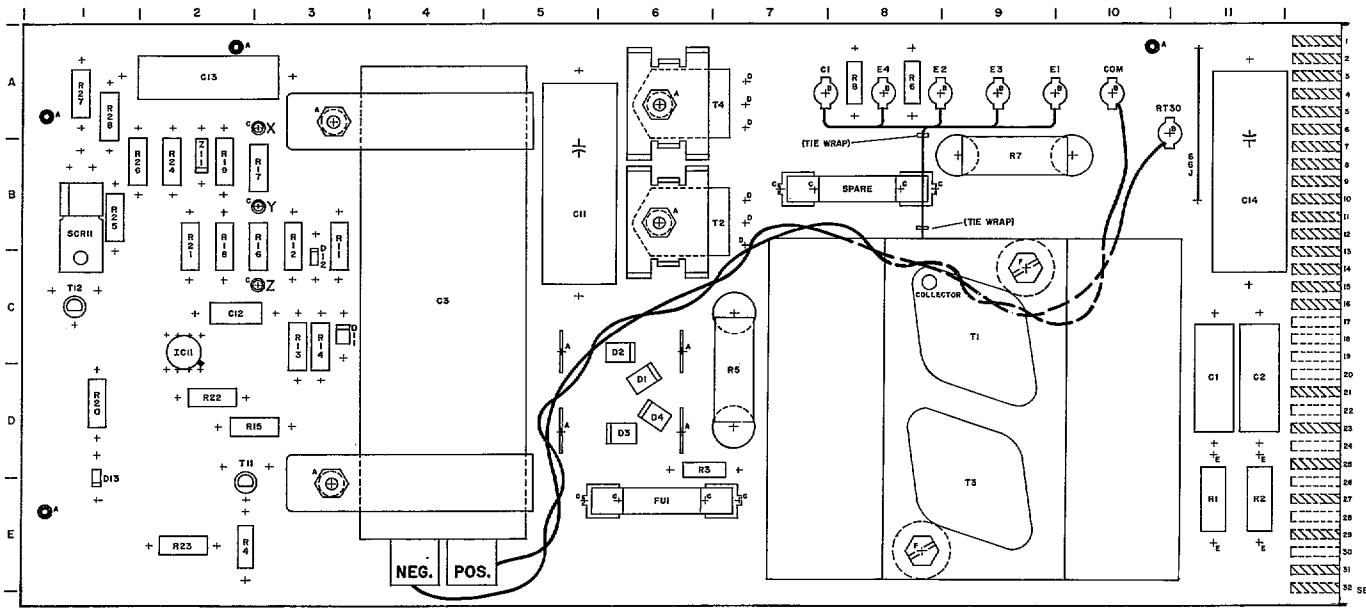
HOLE TABULATION

ALL HOLES ARE .040 EXCEPT THE HOLES TABULATED BELOW.

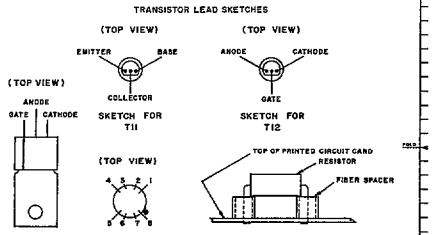
LOC.	DIA.	QTY.
A	.157	12
B	.101	7
C	.070	11
D	.082	6
E	.052	4
F	.189	2

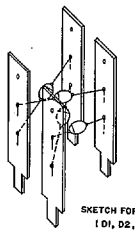
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, 11.50 +.000 −.010 × 5.430 +.002 −.008



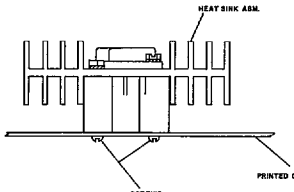
TRANSISTOR LEAD SKETCHES





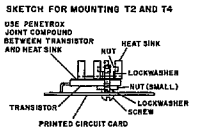
SKETCH FOR DIODE ASM. (D1, D2, D3, & D4)

COUPLE THE FOLLOWING	
WIRE NO.	CONNECTOR
TIE	E1
T1B	E2
T3E	E3
T3B	E4
C	C1
C3−	COM
C3+	RT30



SKETCH FOR MOUNTING TRANSISTOR HEAT SINK ASM.

SKETCH FOR MOUNTING T2 AND T4



2X	36B605266AAD04	
SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.

REVISIONS

No revisions are to be made to this drawing unless authorized in writing by the Product Engineering Section of the Speed Variator Department.

PRINTS TO

5B(2)22	5(S)
5QC2BK	5QC(BK)
X7S150	5LBW(S
AW(BW)	5AE(BW
DS(REP)	5QC(B)

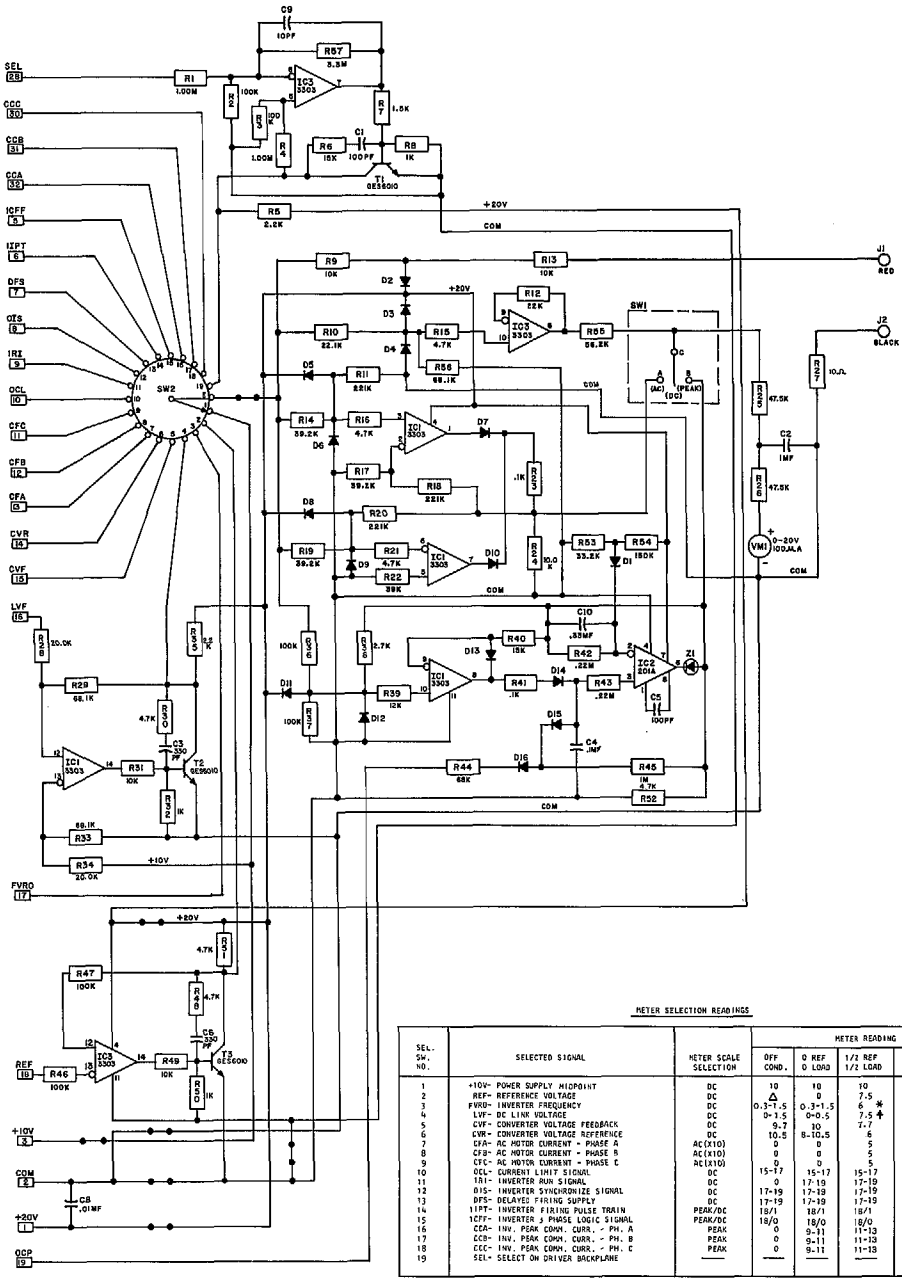
DRAWN J.H. Baker JULY 13, 1976

DESIGN Wm. Mont Feb. 1, 1977

speed variator — ERIE, PA., USA

36D868820AA

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY



METER SELECTION READINGS

SEL. SW. NO.	SELECTED SIGNAL	METER SCALE SELECTION	METER READING				
			OFF COND.	0 REF 0 LOAD	1/2 REF 1/2 LOAD	1 REF 1 LOAD	SPECIAL CONDITION
1	+10V- POWER SUPPLY MIDPOINT	DC	10	10	10	10	10
2	REF- REFERENCE VOLTAGE	DC	△	0	7.5	15	——
3	FVRO- INVERTER FREQUENCY	DC	0.3-1.5	0.3-1.5	6 *	12 *	——
4	LVF- DC LINK VOLTAGE	DC	0-1.5	0-0.5	7.5 †	15 †	——
5	CVF- CONVERTER VOLTAGE FEEDBACK	DC	9.7	10	7.7	5.5	——
6	CVR- CONVERTER VOLTAGE REFERENCE	DC	10.5	8-10.5	6	3.5	——
7	CFA- AC MOTOR CURRENT - PHASE A	AC(X10)	0	0	5	10	15 AT 1.5 LOAD
8	CFB- AC MOTOR CURRENT - PHASE B	AC(X10)	0	0	5	10	15 AT 1.5 LOAD
9	CFC- AC MOTOR CURRENT - PHASE C	AC(X10)	0	0	5	10	15 AT 1.5 LOAD
10	OCL- CURRENT LIMIT SIGNAL	DC	15-17	15-17	15-17	15-17	4-8 AT CURR. LIM.
11	IRI- INVERTER RUN SIGNAL	DC	0	17-19	17-19	17-19	——
12	OIS- INVERTER SYNCHRONIZE SIGNAL	DC	17-19	17-19	17-19	17-19	0 AT SYNCH.
13	DFS- DELAYED FIRING SUPPLY	DC	17-19	17-19	17-19	17-19	——
14	IFPT- INVERTER FIRING PULSE TRAIN	PEAK/DC	18/1	18/1	18/1	18/1	——
15	ICFF- INVERTER 3 PHASE LOGIC SIGNAL	PEAK/DC	18/0	18/0	18/0	18/0	——
16	CCA- INV. PEAK COMM. CURR. - PH. A	PEAK	0	9-11	11-13	12-14	——
17	CCB- INV. PEAK COMM. CURR. - PH. B	PEAK	0	9-11	11-13	12-14	——
18	CCC- INV. PEAK COMM. CURR. - PH. C	PEAK	0	9-11	11-13	12-14	——
19	SEL- SELECT ON DRIVER BACKPLANE		——	——	——	——	——

* READINGS SHOWN ARE FOR 60HZ BASE FREQUENCY SETTING. OTHER READINGS WILL APPLY FOR OTHER BASE FREQUENCIES. ACTUAL FREQUENCIES IN HERTZ IS 5 TIMES METER READING FOR THE 75HZ MAX. FREQUENCY SELECTION AND 10 TIMES THE METER READING FOR THE 150HZ MAX. FREQUENCY SELECTION.

† ACTUAL DC LINK VOLTAGE IS 40 TIMES THE METER READING FOR 460 VOLT DRIVES AND 20 TIMES THE METER READING FOR 230 VOLT DRIVES.

△ DEPENDENT ON REFERENCE SETTING.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

TITLE ELEMENTARY CARD DIAGRAM METER CARD

DRAWN BY [illegible] JUNE 7, 1976

SPEED VARIATION ERIE, PA.

DRAWING NO. 36D86882AA

SHEET 1 OF 2

DASH MOD FOR STANDARD LINE 193X381AAG01

REVISIONS

LTR	DESCRIPTION	DATE	APPROVED
1	[illegible]	5/8/80	[illegible]

DISTRIBUTION

5CB13350 5AJ9415	5CB19A00500BG
AWG(W) 5AA69AW	5CB19A00500BG
DSREP(W)50CB1	5CB19A00500BG

PRINTS TO

36D86882AA

SHEET 1 OF 2

GENERAL ELECTRIC

36D868822AA

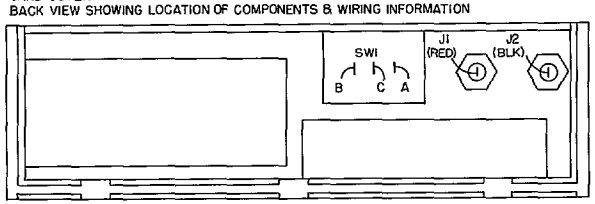
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING — APPLIED PRACTICES | SURFACES ✓ | TOLERANCES (FRACTIONS, DECIMALS, ANGLES)

TITLE PRINTED CIRCUIT DIAGRAM METER CARD

FIRST MADE FOR AF-400

193X381AAG01

CARD COVER
BACK VIEW SHOWING LOCATION OF COMPONENTS & WIRING INFORMATION



HOLE TABULATION

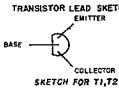
ALL HOLES ARE 040 EXCEPT THE HOLES TABULATED BELOW

LOC	DIA	QTY
A	157	8
B	062	27
C	052	68

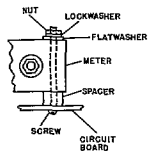
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED
3. CARD SIZE, 11.50 +.000 −.015 × 5.130 +.002 −.008
4. ALL WIRING IS TERMINATED TO THE PRINTED CIRCUIT ON ONE END THE OTHER WIRE TERMINATION IS DESIGNATED BY COMPONENT AND TERMINAL NEAR THE PRINTED CIRCUIT BOARD TERMINATION

TRANSISTOR LEAD SKETCHES



SKETCH FOR T1,T2,T3



SKETCH "A"
MOUNTING OF METER TO PRINTED CIRCUIT BOARD

PRINTED CIRCUIT BOARD
FRONT VIEW SHOWING LOCATION OF COMPONENTS & WIRING INFORMATION

SEE NOTE 1 & 2

SPARE UNIT REQUIREMENT	FRONT	BACK
	2	—
2X	36B605263AA004	—
SCALE	REAR ETCHED BOARD DWG	FRONT ETCHED BOARD DWG

REVISIONS

PRINTS TO

5B(2)22	5(G)		
50C(BK)5	5D(BK)		
5CTS(EO	5L(BW)5		
AW(BW)	5AE(BW)		
DS(REP)	50C(B)		

Erie, PA, USA — speed variator — 36D868822AA

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C764166AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:-

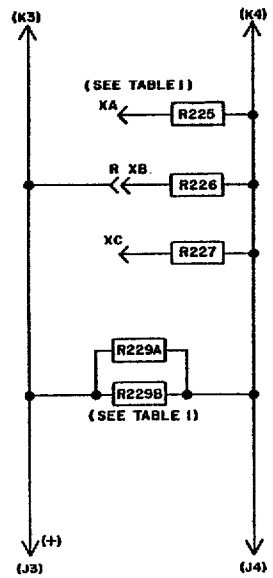
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

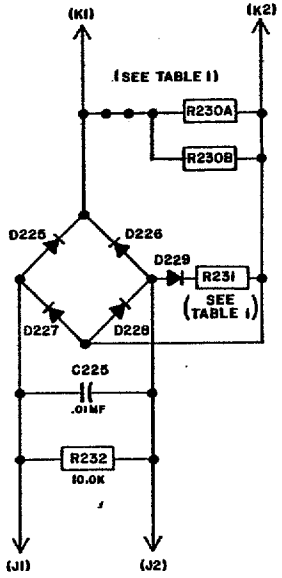
TITLE PRINTED CIRCUIT DIAGRAM
CURRENT FEEDBACK

FIRST MADE FOR AF400

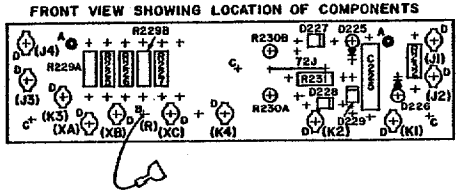
193X382AAG01-G05

SCHEMATIC DIAGRAM





FRONT VIEW SHOWING LOCATION OF COMPONENTS



HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.062	1
C	.187	3
D	.101	11

NOTE: 1. CARD SIZE, 4.900 $^{+.000}_{-.018}$ X 1.300 $^{+.000}_{-.018}$

TABLE 1

GROUP NUMBER	RESISTOR NOMENCLATURE AND VALUE IN OHMS							
	XA R225	XB R226	XC R227	R229A	R229B	R230A	R230B	R231
G01	1210	121	68.1	68.1	182	100	—	562
G02	562	82.5	33.2	27.4	274	100	100	274
G03	392	47.5	18.2	12.1	475	33.2	100	150
G04	681	56.2	27.4	15.0	562	22.1	68.1	100
G05	68.1	27.4	18.2	12.1	150	22.1	27.4	68.1

TABLE 2

GROUP NUMBER	JUMPER SELECTION	RESISTANCE K3 TO K4 OHMS	C.T. TURNS RATIO	NOMINAL INV. RMS AMPS
G01	XA	47.6	2000/1	42
	XB	35.2		56
	XC	28.7		70
G02	XA	23.8	2000/1	84
	XB	19.1		105
	XC	14.2		140
G03	XA	11.45	2000/1	175
	XB	9.45		210
	XC	7.16		280
G04	XA	14.3	4000/1	280
	XB	11.6		350
	XC	9.53		420
G05	XA	9.62	4000/1	420
	XB	7.95		500
	XC	6.93		580

$$\text{VOLTS (K3-K4)} = \frac{\text{INV. RMS AMPS}}{\text{C.T. TURNS RATIO}} \times \text{OHMS (K3-K4)}$$

= 1.0 VOLT RMS NOMINAL

TABLE 3

GROUP NUMBER	RESISTANCE K1 TO K2 OHMS	C.T. TURNS RATIO	COMM. PEAK AMPS
G01	100	2000/1	250
G02	50	2000/1	500
G03	24.9	2000/1	1000
G04	16.7	2000/1	1500
G05	12.2	2000/1	2050

$$\text{VOLTS (K1-K2)} = \frac{\text{COMM. PEAK AMPS}}{\text{C.T. TURNS RATIO}} \times \text{OHMS (K1-K2)}$$

= 12.5 VOLTS PEAK NOMINAL FOR THE POSITIVE COMMUTATION.

NOTE 2: FOR THE NEGATIVE COMMUTATION, A MULTIPLIER OF 0.85 IS APPLIED TO THE ABOVE FORMULA DUE TO THE LOADING EFFECT OF R231.

REV NO. Ø 1

SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT FRONT	BACK
FULL	368G05257AA004		

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1	JUNE 1, 1977 L.R. Baker	CHG. PER. ECA BV-872 #6C	

PRINTS TO

5QC(BK)5	5(S)
5B(2)22	5D(BK)
5L(BW)5	
5(TS)50	
JA(CD)	
AW(BW)	5(QC)B

MADE BY L.R. Baker AUG. 17, 1976
ISSUED L.R. Baker JAN. 19, 1977

APPROVALS cer

speed variator — DIV. OR DEPT.
ERIE, PA. — LOCATION

36C764166AA — CONT ON SHEET — SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

GENERAL ELECTRIC

36C764172AA

SCHEMATIC DIAGRAM

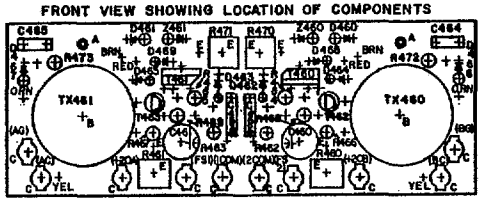
UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
718A515	✓	±	±	±

TITLE PRINTED CIRCUIT DIAGRAM
PULSE TRANSFORMER

FIRST MADE FOR AF-400

193X383AAG01

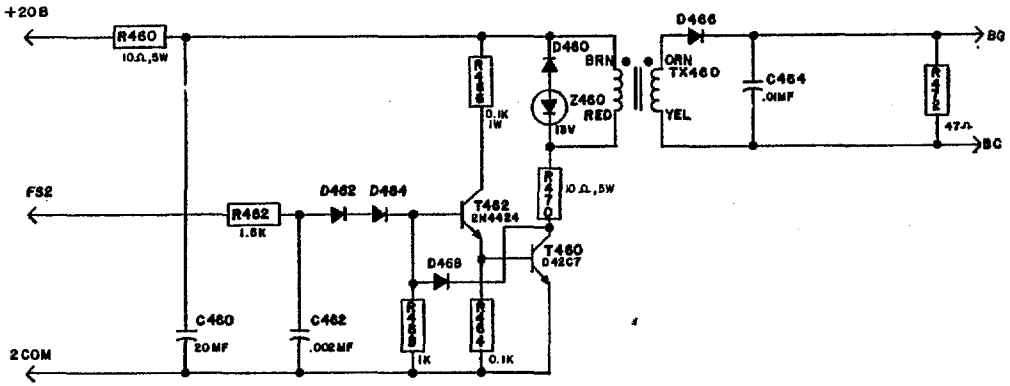
FRONT VIEW SHOWING LOCATION OF COMPONENTS

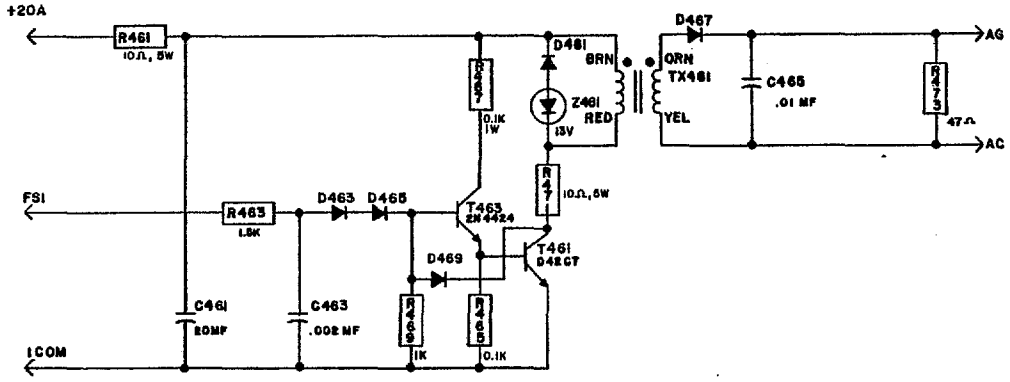


HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

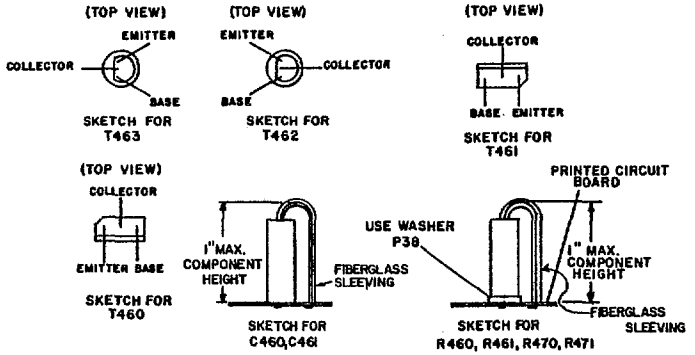
LOC.	DIA.	QUAN.
A	.157	2
B	.187	2
C	.101	10
E	.052	8





NOTE: 1. CARD SIZE 5.130 $^{+.000}_{-.016}$ X 1.90 ±.020

TRANSISTOR LEAD SKETCHES



SCALE	ETCHED CIRCUIT BOARD DWG.	SPACE UNIT REQUIREMENT	
		FRONT	BACK
FULL	368605270AA004	—	—

REVISIONS

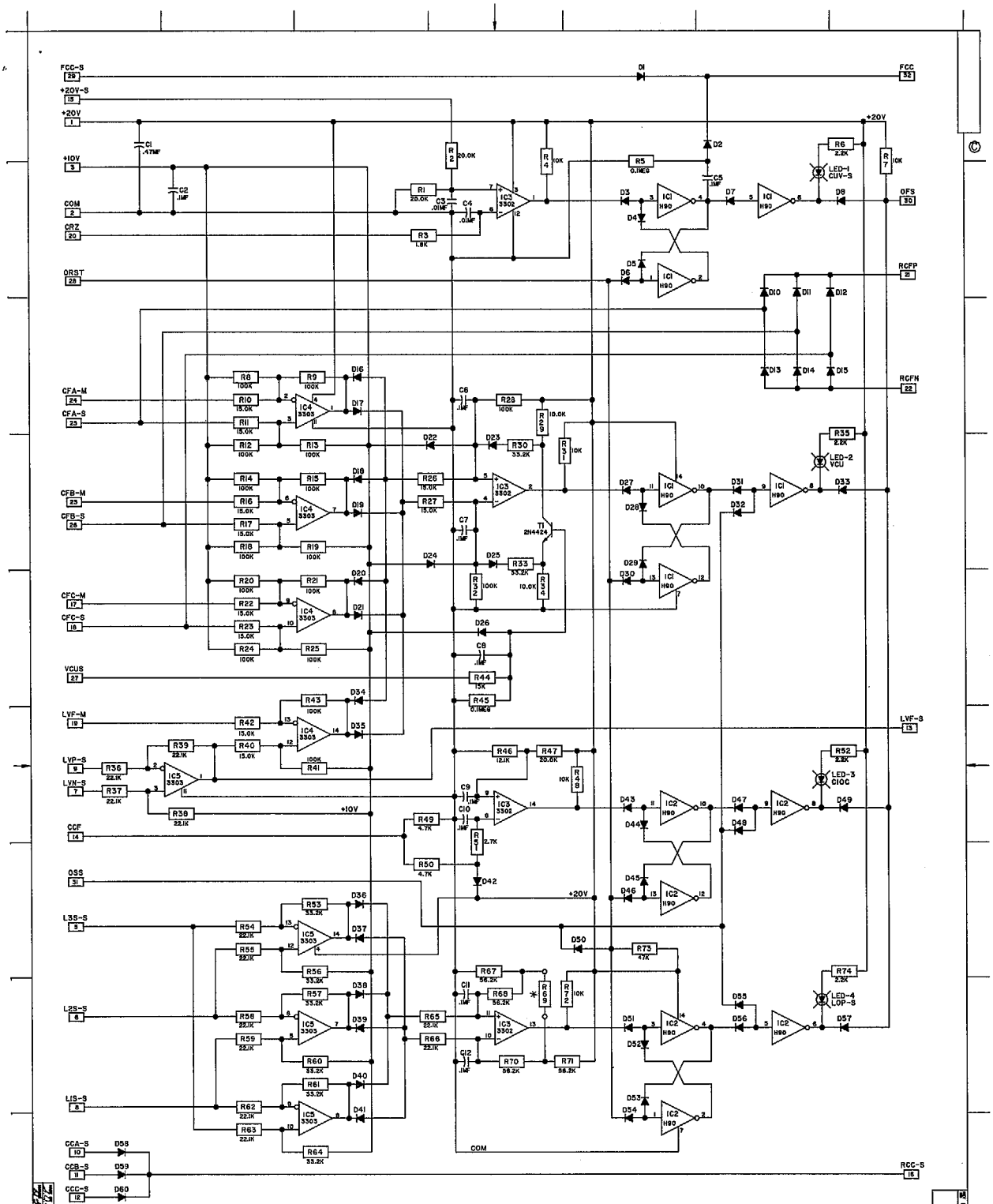
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REV	DESCRIPTION	PER	DATE	APPD
J	Updated per SV672**00			

PRINTS TO

5QC(BK)5	5B(S)
5B(2)22	5D(BK)
5(TS)50	5L(BW)5
(DS(REP)	5AE(BW)
	AW(BW)
	5(QC)8

MADE BY	DATE	APPROVALS	SPEED VARIATOR	DIV OR DEPT	36C764172AA
M S Peyton	JULY 28,1976	ex-gs	ERIE, PA	LOCATION	CONT ON SHEET / SH NO.
ISSUED	Paul Marsh Apr. 1, 1977				



SPEED VARIATOR PROD.			
ERIE, PA.			
36D868831AA			
JACD			
SDCD			
AMEBD			
SUSBW	TSEBD		
SUSBW	SJKBD		
SSBW			

REVISIONS

PRINTS TO

TITLE - ELEMENTARY DIAGRAM
PARALLELING CARD
FIRST MADE FOR AF-400
193X384AASG1

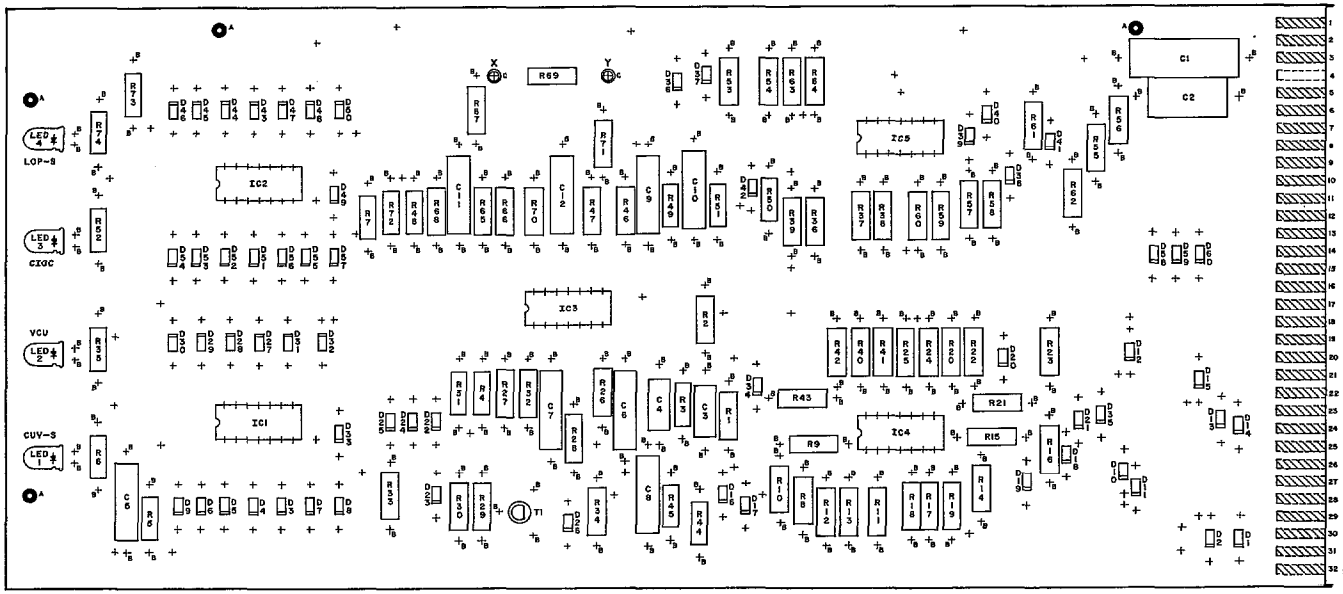
36D868831AA

GENERAL ELECTRIC

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING: —								
APPLIED PRACTICES	SURFACES	TOLERANCE ON MACHINED DIMENSIONS						
		FRACTIONS		DECIMALS		ANGLES		
	✓	±		±		±		

36D868831AA			
CONT ON SHEET FL	SH NO 2		
REV NO 0			
CONT ON SHEET	SH NO.		

TITLE PRINTED CIRCUIT DIAGRAM
PARALLELING CARD
FIRST MADE FOR AF-400 INVERTER
193X384AA601



HOLE TABULATION

ALL HOLES ARE .032 EXCEPT THE HOLES TABULATED BELOW.

LOC.	DIA.	QTY.
A	.157	4
B	.040	161
C	.070	2

NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS.
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, 11.50 +.000 −.015 X 5.130 +.00E −.00S

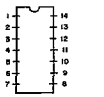
TRANSISTOR LEAD SKETCHES

(TOP VIEW)



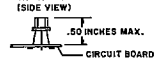
SK. FOR T1

(TOP VIEW)



SKETCH FOR IC1, IC2, IC3, IC4, IC5

COMPONENT HEIGHT SKETCH TRANSISTOR (SIDE VIEW)



1		2X	36B605306AA005	36B605307AA005
FRONT	BACK	SCALE	REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.
SPACE UNIT REQUIREMENT				

REVISIONS

We concur with the changes made to this drawing as noted in the Engineering Section of the Speed Variator Department.

PRINTS TO

SB(B)W	SJ(BW)
SR(2)BW	TS(SO)
SL(2)BW	
AW(BW)	
SD(CO)	
JA(CO)	

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

DRAWN C.R. Baker	OCT. 3, 1977	CAE APPROVALS	speed variator	36D868831AA
TRACED C.R. Baker	APRIL 25, 1978		ERIE, PA., USA	CONT ON SHEET FL SH NO 2